

Persistent Debt and Business Cycles in an Economy with Production Heterogeneity

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ABSTRACT

We study an economy characterized by a time-varying distribution of production to examine the role of debt in amplifying aggregate shocks and propagating recessions. In our model, entrepreneurs utilize risky, long-term debt to finance capital. Liquid assets serve as collateral, and transaction costs make debt illiquid.

Debt payments heighten the volatility of earnings relative to output, deterring entrepreneurs with inadequate collateral from financing efficient levels of capital. This results in a misallocation of resources.

The aggregate effects of risky, illiquid debt held by risk-averse entrepreneurs manifest through cyclical fluctuations in the distribution of production. During recessions, productive entrepreneurs facing substantial debt payments mitigate income risk by deleveraging, thereby reducing their access to capital. As the share of resources used by some of the most productive entrepreneurs decreases, misallocation rises and aggregate total factor productivity declines. The model economy exhibits aggregate nonlinearities that arise from movements in the distribution of entrepreneurs. These movements amplify the fall in TFP over recessions.

The Great Recession involved an unusual fall in the value of business assets. Examining this through the lens of our model, we find a persistent increase in misallocation that generates a slow recovery comparable to the data.

Keywords: Business cycles, Production heterogeneity, Business debt

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1 Introduction

Debt is large in the balance sheets of U.S. firms. Between 1954 and 2006, aggregate non-financial business leverage averaged 35 percent. Thereafter, as business assets declined over the Great Recession, leverage increased by 7 percentage points. The availability of external finance is widely regarded as an important determinant of the efficiency with which resources are allocated. Greater use of debt may signal a more efficient allocation of resources, but high leverage has also been viewed as a force that amplifies adverse shocks and deepens economic downturns.

We focus on a novel channel through which long-term debt affects firm income. Ongoing debt payments increase income volatility, prompting firms to reduce their debt following negative income shocks. As productive assets are diverted to paying off existing debt, income falls further.

This precautionary deleveraging channel carries significant implications for the business cycle. During recessions, firms' efforts to reduce their debt amplify the aggregate shock and intensify the downturn. Heavily indebted yet highly productive firms deleverage the most. As productive firms reduce their leverage, resources are reallocated to less productive entities, resulting in increased misallocation and a decline in aggregate total factor productivity. When a recession involves a substantial decline in the value of business assets, misallocation rises further, slowing the subsequent recovery.

We construct a quantitative framework to analyze the interaction between firm debt and aggregate quantities over the business cycle. The analysis involves an incomplete markets model featuring a time-varying measure of entrepreneurs operating alongside competitive, risk-neutral firms. Entrepreneurs utilize decreasing returns to scale technologies with capital and labor. Capital rentals are subject to collateral constraints, with a producer's financial assets serving as collateral for short-term loans. When an entrepreneur's collateral constraint binds, she may use long-term debt to mitigate its effect. Such loans are costly to adjust, hence illiquid. Overall, the economy is characterized by a distribution of entrepreneurs varying in their idiosyncratic productivity, asset holdings, debt positions and whether they are active, as well as a distribution of workers over labor productivity and assets. Because asset-poor entrepreneurs employ less capital than wealthier counterparts with similar productivity, there is a misallocation of resources. In the aggregate, TFP is endogenous and is determined by the distribution of entrepreneurs across assets and liabilities.

Several novel features of our model distinguish it from existing research on production heterogeneity and debt. In contrast to models with risk-neutral firms (for example, [Khan and Thomas \(2013\)](#)), our entrepreneurs have diminishing marginal utility and savings rates that

are non-linear functions of their assets and debt. Further, efforts to reduce consumption risk motivate productive but asset-poor firms to deleverage during recessions, thereby increasing misallocation. This mechanism is absent in models characterized by one-period debt and risk-neutral firms. Relative to models with risk-averse entrepreneurs, such as [Buera and Shin \(2013\)](#), our framework allows entrepreneurs to hold liabilities, and debt payments play a central role in driving cyclical fluctuations in misallocation.

While we focus on risk-averse firms, our model also includes risk-neutral firms. We assume the latter are large and financially unconstrained. However, risk-averse firms can become as large as, or larger than, risk-neutral firms when they accumulate enough assets.¹ Because there are no frictions among risk-neutral firms, these firms limit the potential impact of financial constraints on the risk-averse firms.

We calibrate the model economy to firm-level real and financial moments. We use moments from *Quarterly Financial Reports* (QFR) computed by [Crouzet and Mehrotra \(2020\)](#), the Small Business Credit Survey (SBCS), and the Flow of Funds from the Board of Governors. We also aim to capture the firm size-over-age distribution, as the financial frictions and access to loans affect how fast firms can grow. For other real variables, we match the employee size distribution of firms. We test our model with untargeted moments from the Kauffman Firm Survey (KFS) and the Canadian Quarterly Survey of Financial Statements (QSFS) and confirm that we have captured critical elements of the link between financial frictions, firm credit, and firm dynamics.

Long-term debt reduces misallocation by allowing productive entrepreneurs to finance a larger scale of production. In its absence there is less entrepreneurship, aggregate output falls 20 percent and TFP by 13 percent. However, holding long-term debt increases income volatility. Interest and principal payments are fixed, while profits vary with idiosyncratic and aggregate shocks. It follows that a negative shock cuts net income by more than it otherwise would. This makes long-term debt a source of misallocation over the business cycle.

Holding long-term debt raises consumption volatility along with income volatility. We illustrate the effects of long-term debt on consumption volatility through the cross-sectional distribution of entrepreneurs' stochastic discount factors (SDF), which measures unequal exposure to consumption risk. In an economy with long-term debt, the distribution of SDFs is more dispersed and right-skewed while in a version without debt, the SDF distribution is left-skewed. More realizations at high values imply that some firms in the economy with long-term debt face a risk of a large fall in their consumption. These entrepreneurs have the

¹Since more than 86 percent of large firms—those with over five hundred employees—are private ([Asker et al. \(2014\)](#)), we do not equate risk-neutral firms with those that employ more than 500 workers. In our calibration, 68% of large firms are entrepreneurs. We also do not assume that non-corporate firms represent all entrepreneurial production.

strongest precautionary motive to cut debt when a recession raises the risk of deeper income losses. We explore this precautionary deleveraging and the misallocation it generates over the business cycle.

A substantial decline in the value of business assets characterized the Great Recession. We analyze the aggregate implications of this phenomenon using our model. During the first eight quarters of the recession, we introduce balance sheet shocks that reduce producers' asset holdings, leading to increased misallocation and a consequent reduction in TFP. An additional, modest negative TFP shock enables the model to jointly replicate both the decline in assets and the aggregate TFP observed in the data.

Even in the absence of shocks to credit supply or the efficiency of financial intermediaries, our model accounts for approximately half of the observed decline in GDP during the Great Recession. Notably, the rise in the dispersion of the marginal revenue product of capital (MRPK), relative to the fall in GDP, aligns with [Alam's \(2020\)](#) empirical work on the cyclical component of MRPK dispersion. This finding corroborates the empirical plausibility of the misallocation channel embedded in our model.

The combination of financial and real shocks results in a protracted recovery, capturing much of the persistence in GDP beyond the trough. If we control for differences in recession severity between the model and the data, the model fully accounts for the slow recovery. This reflects a substantial and persistent reduction in entrepreneurial production following shifts in the distribution of entrepreneurs across assets and debt.

Empirical evidence from the QSFS supports two central predictions of our model. First, using the Whited and Wu index of financial constraints ([Whited and Wu \(2006\)](#)), we find that more productive yet financially constrained firms experience larger reductions in debt and leverage during recessions compared to less productive firms or those less affected by financial constraints. Second, the growth of productive but financially constrained firms is more closely linked to changes in their debt during recessions. This mirrors the model's prediction that highly productive, heavily indebted firms deleverage during downturns, leading to reductions in their output.

Amplification of aggregate shocks, in our model, is a feature of recessions alone. The endogenous response of TFP amplifies the negative shock by 60 percent but does almost nothing in the expansion. Propagation in the recession is the result of a rise in both the dispersion of the marginal revenue product of capital and its skewness, signaling an increase in misallocation. Neither moves in the expansion.

Literature Our work is related to several strands of literature. First, we build on work using models with financial frictions and a time-varying distribution of firms. In contrast to

models of entrepreneurship like [Buera and Shin \(2013\)](#), who focus on economic development, and [Buera and Moll \(2015\)](#), who look at recessions, entrepreneurs in our model have both assets and debt. Compared to business cycle models with risk-neutral firms ([Khan and Thomas \(2013\)](#) and [Jo \(2020\)](#)), we allow for entrepreneurs and risky long-term debt.²

Several papers look at the aggregate effects of firm-level debt in models with default. The majority of this work focuses on short-term debt ([Arellano et al. \(2019\)](#), [Khan et al. \(2021\)](#) and [Ottonello and Winberry \(2020\)](#)).

[Crouzet \(2017\)](#), [Karabarbounis and Macnamara \(2021\)](#), [Jungherr and Schott \(2021\)](#), [Jungherr and Schott \(2022\)](#), and [Poeschl \(2023\)](#) all study heterogeneous firms with long-term debt, but in each the producers are risk-neutral. [Crouzet \(2017\)](#), [Jungherr and Schott \(2021\)](#), and [Poeschl \(2023\)](#) study debt-maturity choice, weighing short-term debt refinancing risk against the debt overhang risk with long-term debt. [Karabarbounis and Macnamara \(2021\)](#) show that a collateral constraint generates misallocation among private firms. Our model also features a collateral constraint that generates misallocation, but our focus is on how this misallocation varies over the business cycle.

[Jungherr and Schott \(2022\)](#) is close to our paper, studying the role of leverage in a business cycle model with long-term debt. They show that risk-neutral firms are slow to reduce leverage at the onset of a recession and this deepens the downturn. Our mechanics are different. In our model risk-averse, productive and levered entrepreneurs deleverage as income risk rises, and this raises misallocation and amplifies the recession.

Among the papers that feature long-term debt, our model shares many elements with [Jeenas \(2026\)](#). He shows that a firm’s holdings of liquid assets are an important predictor of how strongly its investment responds to a monetary policy shock. The relevant cost of investment for a cash-rich firm is not the market interest rate but the forgone return on its liquid assets. This moves little with monetary policy compared to the cost of debt financing. Our environment shares the feature that firms adjust their debt infrequently and that debt is closely tied to investment. In our model, however, the market interest rate also matters for investment as it affects the rate at which entrepreneurs accumulate liquid assets.

Lastly, our work is related to the literature on asymmetries over the business cycle. A body of work has found that contractions are both sharper and shorter than expansions.³ More recently, papers have found cyclicity in higher moments of the distribution of production. Using firm-level panel data, [Salgado et al. \(2019\)](#) show that skewness in the growth rates of employment, sales, and productivity are procyclical. Examining U.S. durable goods

²Seminal papers that explored the role of financial frictions in propagating real shocks include [Bernanke and Gertler \(1989\)](#), [Bernanke et al. \(1999\)](#), and [Kiyotaki and Moore \(1997\)](#).

³See [Neftci \(1984\)](#), [Falk \(1986\)](#), [Hamilton \(1989\)](#), [Sichel \(1993\)](#), [McQueen and Thorley \(1993\)](#), and [Jorda and Taylor \(2016\)](#).

manufacturing at the plant level, [Kehrig \(2015\)](#) finds that the dispersion of TFP is higher in recessions than in booms.

A number of papers study environments leading to asymmetries.⁴ Our work is closest to those that connect aggregate nonlinearities to credit or collateral constraints such as [Kocherlakota et al. \(2000\)](#), [Kuhn and George \(2019\)](#), [Jensen et al. \(2020\)](#), and [Fève et al. \(2021\)](#). These papers study real business cycle models or New Keynesian models, while, in our setting, aggregate quantities depend on a time-varying distribution of firms.

2 Model

2.1 Overview

The economy consists of a continuum of infinitely-lived households, risk-neutral firms, banks, and a government. There are two types of households, workers and entrepreneurs, differentiated by their ability to organize production; only entrepreneurs can operate a firm. Workers are indexed by their holdings of liquid assets a and their idiosyncratic labor productivity ε . They are subject to uninsurable idiosyncratic shocks to their labor productivity and supply labor inelastically to competitive entrepreneurs. The latter are indexed by their holdings of liquid assets a , long-term debt b , and their idiosyncratic productivity z , which is stochastic. Active entrepreneurs operate a firm. Inactive entrepreneurs do not have a business; instead, they supply labor for a wage. Depending on their choice, z is their labor productivity or the total factor productivity of their firm. When an entrepreneur chooses to start a new business, she has to pay an entry cost, $\kappa(z)$. An entrepreneur can borrow up to $\gamma_k \kappa(z)$ to help finance entry costs.

Entrepreneurs operate firms by renting capital and hiring labor in spot markets. Capital is rented subject to a collateral constraint, and liquid assets serve as collateral. While all entrepreneurs can rent capital, those operating startups are initially unable to borrow. Access to external finance arrives randomly to those who cannot presently borrow. Those who borrow may also experience idiosyncratic shocks that temporarily shut down their access to new loans.

All borrowing is in the form of long-term debt, b , which is defaultable. Default results in a record of bankruptcy in an entrepreneur's credit history. Such bankruptcy flags prevent entrepreneurs from further borrowing and operating a business. Beyond default, active entrepreneurs may choose to exit or be forced to do so after receiving an exit shock. Inactive

⁴Adjustment costs are explored by [McKay and Reis \(2008\)](#) and [Jovanovic \(2006\)](#). Learning is studied by [Chalkley and Lee \(1998\)](#), [Van Nieuwerburgh and Veldkamp \(2006\)](#) and [Acemoglu and Scott \(1997\)](#).

entrepreneurs without default in their credit history may choose to enter production and become active.

A measure of risk-neutral firms produces goods alongside active entrepreneurs. However, these firms are not subject to idiosyncratic productivity shocks nor collateral constraints on renting capital. Thus, they produce efficiently. They return profits as dividends to households. We assume that all workers and entrepreneurs hold the same quantity of shares in these risk-neutral firms. These firms operate continuously; they do not enter or exit.

Banks price debt to reflect entrepreneurs' default risks. The financial sector is competitive, and banks expect zero profits on each loan. The government collects taxes from households; tax revenues finance transfers and other government spending.

The aggregate state $\Omega = (g, A)$ consists of the household distribution g over $(a, b, \varepsilon, z, o)$ and the exogenous component of aggregate total factor productivity, A . The status indicator o applies to entrepreneurs and takes four values: active with borrowing access ($o = 1$), active without borrowing access ($o = 2$), inactive ($o = 3$), and inactive with a default flag ($o = 4$). We denote the distributions over workers and entrepreneurs by g^w and g^e . Time is continuous.

2.2 Households

2.2.1 Worker

Each worker's labor productivity follows a Poisson process. Shocks occur with frequency λ_ε and result in a new productivity drawn from a time-invariant distribution. Labor productivity shocks arrive independently across workers, and one worker's productivity is independent of others. Households receive utility flow from consuming non-durable goods, c , and their utility function is

$$u(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0.$$

As worker-type households cannot start a business, they solve the following problem.

$$\begin{aligned} \rho v^w(a, \varepsilon; \Omega) &= \max_c u(c) + \frac{\partial v(a, \varepsilon; \Omega)}{\partial a} \dot{a} + \sum_{\varepsilon' \neq \varepsilon} \lambda_\varepsilon^{\varepsilon \rightarrow \varepsilon'} [v^w(a, \varepsilon'; \Omega) - v^w(a, \varepsilon; \Omega)] \\ \dot{a} &= \varepsilon w(\Omega) + r(\Omega)a - c - T(a, \varepsilon, \Omega) + d(\Omega), \quad a \geq \underline{a}. \end{aligned} \tag{1}$$

Above, $w(\Omega)$ is the real wage, the real interest is $r(\Omega)$ and $d(\Omega)$ is the dividend received from risk-neutral firms. Government taxes, $T(a, \varepsilon, \Omega)$, are a non-linear function of assets and labor income.

2.2.2 Entrepreneur

Productivity Entrepreneur-type households have idiosyncratic total factor productivity, z , which follows a Poisson process. Similar to workers, entrepreneurs receive a productivity shock with frequency λ_z . This entails a draw of new productivity from a time-invariant distribution. Active entrepreneurs' productivity is also tied to their ability to borrow. When they cannot increase debt ($o = 2$), their productivity is ψz where $\psi < 1$.

Assets An entrepreneur's liquid assets, a , serve as collateral against rented capital, k ; $k \leq \gamma a$. In the absence of the collateral constraint, capital chosen by an entrepreneur would only depend on her productivity and input prices. However, the collateral requirement distorts active entrepreneurs' capital decisions and provides an incentive to take on debt to raise liquid assets.

Debt Debt is long-term and defaultable. Entrepreneurial households holding debt b pay interest and a fraction of the principal each instant. Those active entrepreneurs able to take on new loans, ($o = 1$), may increase their debt by paying an adjustment cost, up to $\underline{b}$. Due to default risk, lenders price the loan through a discount function $q(a', b', z, o; \Omega)$, which varies with b' and the borrower's state. Households that may reduce their debt, but cannot borrow more, are indexed by $o = 2$. Both borrowing and prepayment carry an adjustment cost of $\xi_b |b'| + \xi_{b0}$, where b' is the debt chosen at adjustment.

When an entrepreneur chooses to default, she must exit and absorb a utility cost, ξ_d . Her debt is erased and she is able to keep a fraction of assets, χa . Moreover, she will have default recorded in her credit history, which excludes her from borrowing or re-entry into production. For tractability, we assume that this default flag is removed stochastically with intensity λ_d .

Entrants are able to borrow to finance a fraction of their entry cost. Thereafter, they begin production but cannot take additional loans ($o = 2$). The option to borrow arrives randomly with the rate λ_{21} . Any entrepreneur able to borrow may lose their option at the rate λ_{12} .

Entrepreneurs' problem

In summary, active entrepreneurs, with $o \in [1, 2]$ produce, and those of type $o = 1$ can also increase or decrease b , while those with $o = 2$ can only decrease b . An inactive entrepreneur without a bankruptcy flag ($o = 3$) can start a business. We summarize individual states as $\omega = (a, b, z, o)$. An entrepreneur solves the following problem,

$$\begin{aligned}
v(\omega; \Omega) &= \max_{\{c_t, (k_t, l_t) | o \in [1, 2]\}_{t \geq 0, \tau}} \left\{ E_0 \int_0^\tau u(c_t) dt + e^{-\rho\tau} v^*(\omega_\tau; \Omega_\tau) \right\}, \\
v^*(\omega_\tau; \Omega_\tau) &= \begin{cases} \max\{v^r, v^x, v^b\} & \text{if } o = 1 \\ \max\{v^p, v^x, v^b\} & \text{if } o = 2 \\ v^e & \text{if } o = 3, \end{cases} \quad (2)
\end{aligned}$$

ω is given.

Above, an entrepreneur chooses non-durable consumption $\{c_t\}$. Active entrepreneurs also choose capital $\{k_t\}$ and labor $\{l_t\}$, and produce output using a decreasing-returns-to-scale technology, $y_t = A_t z_t k_t^\alpha l_t^\nu$. Entrepreneurs without default flags choose their optimal stopping time τ . At the stopping date, a household makes a discrete choice. Conditional on the entrepreneur's current status, o , she may choose (i) a large shift in asset position, (ii) a change in occupation or (iii) default. When an entrepreneur chooses to stop, she chooses one of the following options: i) refinancing debt, ii) prepaying debt, iii) exiting production after repaying debt, iv) defaulting on b and exiting, and v) starting a business. The values corresponding to these choices are v^r , v^p , v^x , v^b and v^e , respectively and defined below in (4).

The Hamilton-Jacobi-Bellman (HJB) equation corresponding to active entrepreneurs' problems, prior to stopping is,

$$\begin{aligned}
\rho v(\omega; \Omega) &= \max_{c, (k, l)_{o \in [1, 2]}} u(c) + \frac{\partial v(\omega; \Omega)}{\partial a} \dot{a} + \sum_{z' \neq z} \lambda_z^{z \rightarrow z'} [v(a, b, z', o; \Omega) - v(a, b, z, o; \Omega)] \\
&+ \mathbf{1}_{o \in [1, 2]} \left(\frac{\partial v(\omega; \Omega)}{\partial b} \dot{b} - \lambda_{10} [v(a, b, z, o; \Omega) - v(a - b - \xi_{b0}, 0, z, 3; \Omega)] \right. \\
&\left. - \lambda_b [v(a, b, z, o; \Omega) - v(a, b, z, o'; \Omega)] \right) - \mathbf{1}_{o=4} \lambda_d [v(a, b, z, 4; \Omega) - v(a, b, z, 3; \Omega)] \\
\dot{a} &= \begin{cases} \pi(z(o), \Omega) + r(\Omega)a - c - (r(\Omega) + \theta)b - T(z, a, b, \Omega) + d(\Omega), & k \leq \gamma a \text{ if } o \in [1, 2] \\ zw(\Omega) + r(\Omega)a - c - T(z, a, \Omega) + d(\Omega) & \text{if } o \in [3, 4] \end{cases} \\
\dot{b} &= -\theta b \\
\pi(z(o), \Omega) &= Az(o)k^\alpha l^\nu - w(\Omega)l - (r(\Omega) + \delta)k - \xi(z) \\
v(\omega; \Omega) &\geq v^*(\omega; \Omega) \quad (3)
\end{aligned}$$

where $z(o)$ is an active entrepreneur's effective productivity, equal to z when $o = 1$ and ψz

when $o = 2$; $\xi(z)$ is a fixed operating cost that varies with productivity; and $v^*(\omega; \Omega)$ is the stopping value, the maximum over the stopping choices defined in (4).

Beyond productivity shocks, entrepreneurs face exogenous exit shocks which makes them inactive. When exiting, they need to pay the existing debt and the debt adjusting cost, their assets after exiting will be $a - b - \xi_{b0}$. Thus, the expected change in value, for an active entrepreneur experiencing an exit shock, is $\lambda_{10}[v(a, b, z, o; \Omega) - v(a - b - \xi_{b0}, 0, z, 3; \Omega)]$. The last term of the HJB equation, $\lambda_b[v(a, b, z, o; \Omega) - v(a, b, z, o'; \Omega)]$ reflects the expected change in value from a change in the access to new loans. The transition rate λ_b and the switching state o' depend on the current borrowing status: when $o = 1$, the entrepreneur transitions to $o' = 2$ at rate $\lambda_b = \lambda_{12}$, and when $o = 2$, she transitions to $o' = 1$ at rate $\lambda_b = \lambda_{21}$.

The stopping value $v^*(a_\tau, b_\tau, z_\tau, o_\tau; \Omega)$ in the last constraint of (3) is the maximum of the available options depending on o , all of which are listed below. These options define the value from adjusting an existing loan, exiting or defaulting.⁵

1. Refinancing b , ($o_\tau = 1$): $v^r(a_\tau, b_\tau, z_\tau, o_\tau; \Omega) = \max_{a', b'} v(a', b', z_\tau, o_\tau; \Omega)$

$$a' = a_\tau - b_\tau + q(a', b', z_\tau, o_\tau; \Omega)b' - \xi_b|b'| - \xi_{b0}, \quad \underline{b} \geq b'$$
2. Prepaying b , ($o_\tau = 2$): $v^p(a_\tau, b_\tau, z_\tau, o_\tau; \Omega) = \max_{a', b'} v(a', b', z_\tau, o_\tau; \Omega)$

$$a' = a_\tau - b_\tau + q(a', b', z_\tau, o_\tau; \Omega)b' - \xi_b|b'| - \epsilon|b' - b| - \xi_{b0}, \quad b_\tau \geq b'$$
3. Exiting, ($o_\tau = 1, 2$): $v^x(a_\tau, b_\tau, z_\tau, o_\tau; \Omega) = v(a - b - \xi_{b0}, 0, z, 3; \Omega)$
4. Defaulting on b , ($o_\tau = 1, 2$): $v^b(a_\tau, b_\tau, z_\tau, o_\tau; \Omega) = v(\chi a, 0, z, 4; \Omega) - \xi_d$
5. Entering, ($o_\tau = 3$): $v^e(a_\tau, 0, z_\tau, 3; \Omega) = \max_{a', b'} v(a', b', z_\tau, 2; \Omega) - \xi_e$

$$a' = a_\tau + q(a', b', z_\tau, 2; \Omega)b' - \xi_b|b'| - \xi_{b0} - \kappa(z), \quad b' \leq \gamma_k \kappa(z)$$
(4)

2.3 Risk-neutral firms

Risk-neutral firms produce goods using a decreasing returns to scale technology with the same factor shares as entrepreneurs' technology. Nonetheless, they are not subject to a collateral constraint on the capital they rent, nor do they experience idiosyncratic productivity shocks.

Similarly to entrepreneurs, risk-neutral firms hire labor after observing total factor productivity. They also pay fixed operating costs and distribute after-tax dividends to house-

⁵When $o = 2$ and the entrepreneur chooses to reduce their debt, we add a small cost proportional to the size of the adjustment ($\epsilon|b' - b|$), where $\epsilon = 0.005$, for numerical stability of the solution algorithm.

holds, at tax rate τ_L . Below is their problem.

$$r(\Omega)v^L(\Omega) = \max_{k_L, l_L} (1 - \tau_L) [Az_L k_L^\alpha l_L^{1-\alpha} - w(\Omega)l_L - (r(\Omega) + \delta)k_L - \xi(z_L)] \quad (5)$$

In the absence of financial constraints, these producers have no reason to accumulate assets or use debt. Below, we map such firms to large firms in the data, thereby fitting the size distribution. Since we abstract from financial constraints faced by these firms, their presence limits the aggregate effects of debt held by entrepreneurs.

2.4 Financial intermediaries

There are risk-neutral, competitive financial intermediaries. These intermediaries issue short-term deposits and lend to active entrepreneurs. Given the risk of default on loans, banks offer loan rates based on borrowers' portfolios of assets and debt as well as their income. They earn zero expected profits from each loan.

A borrower's loan rate involves two components. The first is a premium for the probability of default which varies with both individual and aggregate states. The second is the stream of risk-free real rates over the expected duration of the loan, $r_{t,t \in [0, \tau]}$. The risk premium is embodied in a discount factor, $q(\omega_0; \Omega_0)$ and a borrower with initial debt b_0 , receives $q(\omega_0; \Omega_0)b_0$. Each instant thereafter, she has to pay the risk free real interest rate r_t and a fraction θ of the remaining balance b_t . Therefore, the flow income from a loan is $(r_t + \theta)b_t$. It follows that a financial intermediary's discount rate for a loan is $r_t + \theta$ as the loan matures at the rate θ .

Since the intermediaries earn zero expected profit on each loan, the discounted value of a loan at origination has to be equal to the expected cash flow from it. Hence, the price of the loan region is given by

$$q(\omega_0; \Omega_0)b_0 = \mathbb{E} \int_0^\tau e^{-\int_0^s (r_s + \theta) ds} (r_t + \theta)b_0 dt + e^{-\int_0^\tau r_s ds} b(\omega_\tau; \Omega).$$

This pricing equation only applies for individual states in the non-default region as a borrower who would default immediately will not receive a loan.

Recall that when an entrepreneur defaults, she retains χa of her assets; intermediaries recover $\min[(1 - \chi)a, b]$. Applying the Feynman-Kac formula, the above equation can be

written as the following partial differential equation,

$$\begin{aligned}
& (\theta + r(\Omega))q(\omega; \Omega) = \theta + r(\Omega) + \frac{\partial q(\omega; \Omega)}{\partial a} \dot{a} + \frac{\partial q(\omega; \Omega)}{\partial b} \dot{b} \\
& + \sum_{z' \neq z} \lambda_z^{z \rightarrow z'} [q(a, b, z', o; \Omega) - q(a, b, z, o; \Omega)] \\
& - \lambda_{10} [q(a, b, z, o; \Omega) - q(a - b - \xi_{b0}, 0, z, 3; \Omega)] \\
& - \lambda_b [q(a, b, z, o; \Omega) - q(a, b, z, o'; \Omega)] \quad \text{for } t \in [0, \tau), \\
& q(\omega; \Omega) = \frac{\min[(1 - \chi)a, b]}{b} \quad \text{for } t = \tau.
\end{aligned} \tag{6}$$

These loan prices only exist for active entrepreneurs, $o \in [1, 2]$. As in active entrepreneurs' problems, the transition rate λ_b and the switching state o' depend on o : when $o = 1$, the entrepreneur transitions to $o' = 2$ at rate $\lambda_b = \lambda_{12}$, and when $o = 2$, she transitions to $o' = 1$ at rate $\lambda_b = \lambda_{21}$.

2.5 Government

The government collects taxes from households and risk-neutral firms. Taxable income includes labor income of workers and inactive entrepreneurs supplying labor, and profits of active entrepreneurs, as well as interest and dividend income of all households. Active entrepreneurs may deduct loan interest payments from their taxable income. Tax revenues are spent on government consumption which is not valued by households.

When necessary, the government can also hold households' assets. It takes on the corresponding capital and rents it to the capital market at the prevailing rental rate, and rebates the rental income lump-sum to all households.

2.6 Equilibrium

A recursive competitive equilibrium is a set of functions, solving workers, entrepreneurs, risk-neutral firms, and financial intermediaries problems and clearing the markets for assets, debt, labor and output. It is described by the following conditions.

1. Households optimize. Given prices $\{r, w, q\}$, v^w solves (1), v solves (3) subject to (4).
2. Risk-neutral firms maximize profits by solving (5).
3. Competitive financial intermediation implies the debt discount rate q is determined by (6).
4. Asset market clears: $\int a g^w(a, \varepsilon) d[a \times \varepsilon] + \int (a + b) g^e(a, b, z, o) d[a \times b \times z \times o] = \int_{o \in [1, 2]} k g^e(a, b, z, o) d[a \times b \times z \times o] + k_L$.

5. Labor market clears: $\int \varepsilon g^w(a, \varepsilon) d[a \times \varepsilon] + \int_{o \in [3,4]} z g^e(a, b, z, o) d[a \times b \times z \times o]$
 $= \int_{o \in [1,2]} l g^e(a, b, z, o) d[a \times b \times z \times o] + l_L.$
6. The government budget constraint holds.

The asset market clearing condition in 4 equates the net assets held by workers and entrepreneurs with the aggregate capital stock, over time. In 5, the aggregate supply of labor is the sum of labor supplied by workers and inactive entrepreneurs. While the former is constant, entrepreneurs' labor supply varies with entry and exit decisions.

3 Mapping Model to Data

Aggregate results from our model are from financial frictions involving long-term debt held by risk-averse firms. Furthermore, firms' investment and growth are closely linked to their use of debt. Using data to discipline the mechanics of this linkage requires access to the balance sheets of U.S. firms. However, the vast majority of firms are privately held and, for private firms, there is no widely used data source comparable to Compustat. Given this, we assemble our moments used in calibration from the U.S. Census Bureau's Quarterly Financial Report, the Small Business Credit Survey, and the Flow of Funds. Next, we check the resulting quantitative model against untargeted moments from the Kauffman Firm Survey and the Canadian Quarterly Survey of Financial Statements (QSFS). Throughout, we aim to match not just how much firms borrow on average, but how debt and leverage are distributed across them.

Firm dynamics are also important to this discipline: the severity of the collateral constraint governs the rate at which firms grow. To pin down the effect of debt on aggregate quantities, we therefore also match the size distribution of firms and the size-by-age distribution.

We assign values for a subset of parameters before solving the model's steady state. These are either parameters that can be linked to data values independently of the model's solution or parameters that have commonly used values. In Table 1 they are distinguished with N in the column labelled 'Internal'. The remaining 21 parameters, designated Y in the internal column of the table, as well as the share of entrepreneur type in the population are jointly calibrated. Overall, the number of parameters is 22 and the number of targets is 27. Table 1 lists calibrated parameters, and Table 3 reports targeted data moments and model moments. In the following, we associate certain targets with specific parameters, but since parameters are jointly determined, these associations are approximate and not exact.

Table 1: Parameter values

Parameter	Value	Internal	Description
Preference			
ρ	0.078	N	Discount rate
σ	3.0	N	Curvature of the utility function
Tax			
τ_0	1.14	Y	Individual tax parameter
τ_1	0.181	N	Individual tax rate progressivity
τ_L	0.21	N	Tax rate for risk-neutral firms
Labor productivity			
$\bar{\varepsilon}$	12.5	N	Mean of worker productivity
η_ε	1.2	Y	Shape of Pareto distribution
λ_ε	0.1	Y	Shock intensity
Production productivity			
$\bar{z}$	0.905	N	Mean of entrepreneurial productivity
η_z	2.13	Y	Shape of Pareto distribution
λ_z	0.1	Y	Shock intensity for entrepreneurs
ψ	0.75	Y	Productivity shock when $o = 2$
z_L	1.22	Y	Risk-neutral firm productivity
Production			
α	0.285	N	Capital share
ν	0.6	N	Labor share
δ	0.07	N	Depreciation rate
$\kappa(z)$	$\{-2.5, 0.0\}$	Y	Entry cost bounds
ξ_e	0.2	Y	Entry cost parameter
ξ	0.82	Y	Production cost parameter
λ_{12}	0.2	Y	Borrowing option shock frequency, $o = 1 \rightarrow 2$
λ_{21}	0.25	Y	Borrowing option shock frequency, $o = 2 \rightarrow 1$
λ_{10}	0.07	Y	Exit shock frequency
Assets and debts			
θ	0.06	Y	Amortization rate of debt
γ	1.0	Y	Collateral constraint parameter
γ_k	0.9	Y	Borrowing constraint of entering cost
ξ_b	0.03	Y	Loan origination cost parameter 1
ξ_{b0}	0.01	Y	Loan origination cost parameter 2
ξ_d	0.15	Y	Default cost parameter
$\underline{b}$	440.0	Y	Debt limit
χ	0.25	Y	Retention rate under default
λ_d	0.1667	N	Shock intensity for removal of default flag

Preferences We assume iso-elastic utility and set the coefficient of relative risk aversion, σ , to 3. The discount rate ρ is set to 0.078.

Tax The income tax function $T(y) = y - \tau_0 y^{1-\tau_1}$ is taken from [Heathcote et al. \(2017\)](#) where y is taxable income. Taxable income is labor income (wage workers and inactive entrepreneurs) and profits (active entrepreneurs) plus interest and dividend income. The parameter τ_1 , determining the degree of progressivity of the tax system, is 0.181 as in [Heathcote et al. \(2017\)](#).⁶ Next, τ_0 is set to 1.14 to match a tax revenue-output ratio of 16.7%.⁷ The tax function implies transfers when taxable income is low and taxes at higher levels of income. We assume the government consumes revenues net of transfers.

Productivity Both labor and entrepreneurial productivity are assumed to be drawn from Pareto distributions. Each instant, a shock may arrive and an individual receives a new productivity draw. To specify this process, we need to choose values for the curvature parameters of the Pareto distributions $(\eta_\varepsilon, \eta_z)$ and shock intensities $(\lambda_\varepsilon, \lambda_z)$.⁸

Parameter values for the idiosyncratic total factor productivity process of entrepreneurs are important in shaping the firm size distribution. We choose the mean of discretized z , $\bar{z}$, to be 0.905. When active entrepreneurs lose their ability to increase debt, they also experience a negative productivity shock. We assume that their productivity falls by 25%.⁹

We set the productivity of risk-neutral firms, z_L , to 1.22. We assume these are large firms that each employ at least 500 workers. While the majority of entrepreneurs are smaller than risk-neutral firms, we allow some to employ more than 500 workers.¹⁰ Thus, we choose the mass of risk-neutral firms to account for 0.35% of all firms in the steady state. This implies they are 87.5 percent of firms with more than 500 employees and produce 52 percent of GDP.

We set the parameters for labor productivity to match the earnings distribution (Survey of Consumer Finances, SCF). Since entrepreneurs' business incomes also shape the earnings distribution, their idiosyncratic total factor productivity also helps determine the earnings distribution. Table 2 shows the data and model fit.

⁶They estimate this parameter using the Panel Study of Income Dynamics (PSID) for survey years 2000, 2002, 2004, and 2006, in combination with the NBER's TAXSIM program.

⁷We use the tax revenue-output ratio between 2000 and 2014 reported by the Congressional Budget Office.

⁸We discretize the support of entrepreneurs productivity distribution into 3 grid points that are not linearly spaced. Instead, the 3 points are chosen to capture the bottom 75.0, 20.0, and 5.0 percentiles of the population of entrepreneur-type households. In contrast, the labor productivity grid is equal-spaced with 10 points.

⁹As young firms are initially unable to borrow, this helps slow down their growth. For firms with debt, this raises the risk of default.

¹⁰To convert the model's labor into employees, we multiply each entrepreneur's labor by $\frac{500}{l_L}$.

Table 2: Earnings distribution

	Quintiles(%)					Top(%)		
	1q	2q	3q	4q	5q	90-95	95-99	99-100
Data	-0.1	3.5	11.0	20.6	65.0	12.1	18.3	18.0
Model	2.7	9.3	15.1	20.3	52.6	11.8	11.0	14.4

Data: SCF (2010)

Production We set labor’s share, in the production function, ν , to 0.6 and the annual depreciation rate δ to 0.07. These values are consistent with those in [Khan and Thomas \(2013\)](#).

Entry costs rise with productivity as they are proportional to unconstrained steady state profits (flow profits of a firm with a non-binding collateral constraint). Specifically, $\kappa(z) = e^{\kappa_z} \pi(z, g^*)$ where $\pi(z, g^*)$ is an unconstrained firm’s profit in the steady state (g^*) and κ_z is an evenly spaced vector corresponding to the values of z . We set the lower bound of κ_z to -2.5 and the upper bound to 0.0.¹¹ Having entry costs that rise with productivity helps match the smaller relative size of young firms, those aged 0–5 and 6–10 years. In addition, all entrants pay a common utility cost of entry, ξ_e , which we set to 0.2.

The operating cost, $\xi(z)$, also rises with total factor productivity, $\xi(z) = \xi z^{\frac{1}{1-\alpha-\nu}}$. Setting higher fixed costs for more productive entrepreneurs reduces their earnings. This slows wealth accumulation, which helps us better match the growth rate of young firms.

The borrowing option shock frequencies, λ_{12} and λ_{21} , are set to 0.2 and 0.25, respectively, implying average durations of approximately 5 years with access to new loans and 4 years without. This shock is essential for capturing the relative size of young firms since cash-poor firms have limited ability to grow when they cannot borrow. (Recall we assume that after initially being able to finance 90% of their entry cost, entrants start in the no borrowing state, $o = 2$.) The rate at which entrants gain access to loans affects both the value of entry and the speed at which firms grow.¹²

The frequency of exit shocks λ_{10} is 0.07, implying that a firm experiences this shock on average every 14 years. Firms in our model exit as a result of this shock, and also by choosing to default. Given the latter, the shock frequency is set to match the exiting rate of 8.5% (BDS).

We set 70% of the population as workers and 30% as entrepreneur types. In the steady

¹¹As an example, when there are 3 values of z , the vector $[\kappa_{z1}, \kappa_{z2}, \kappa_{z3}]$ will be $[-2.5, -1.25, 0]$. This leads to an entry cost, for an entrepreneur with productivity z_1 , of $e^{-2.5} \pi(z_1, g^*)$.

¹²The no-borrowing shock is consistent with the observation that not all firms have credit access at all times. The Small Business Credit Survey reports that, on average, 47% of firms received the full loan amount they applied for over 2016–2019. In the steady state of our model, 46.7% of active entrepreneurs have $o = 1$ and are able to increase debt.

state, 34.5% of entrepreneurs are active which implies that 10.4% of the population engages in entrepreneurship. This falls into the range found by [Cagetti and De Nardi \(2006\)](#) who, using the SCF, showed that the share of entrepreneurs ranged from 7.6%-16.7%.

Assets and debts Our target moments on the debt distribution are from three sources. The first is [Crouzet and Mehrotra \(2020\)](#), who draw on the Quarterly Financial Report (QFR) by the US Census Bureau. The QFR surveys firms across manufacturing, mining, wholesale trade, and retail trade, and crucially covers both public and private corporations—unlike Compustat. From this source, we target debt-to-asset ratios as a function of firm size as measured by the value of their assets.

The second source is the SBCS, a national sample of firms with fewer than 500 employees. We match two moments from this survey: the share of small firms carrying no debt, and the share with outstanding debt between \$1 and \$1,000,000.^{13,14} The third source is the Flow of Funds, from which we target the aggregate leverage ratio of non-financial businesses.

The following parameters play the largest role in matching the moments listed above. The loan origination cost, ξ_b is set to 3% of the size of a new loan. There is also a fixed cost, ξ_{b0} , set to 0.01. The amortization rate of debt, θ is set to 6% which implies a half-life of approximately 12 years if entrepreneurs do not refinance or prepay the loan.¹⁵ While the loan price schedule for long-term debt b imposes an endogenous borrowing limit, we also set an exogenous borrowing limit, $\underline{b}$. This helps match the distribution of firm leverage.

The collateral constraint parameter γ is set to 1.0 and the fraction of entry costs that must be self financed, $1 - \gamma_k$, is 10 percent. Together with the frequencies of moving between $o = 1$ and $o = 2$, λ_{12} and λ_{21} , these parameters help us match the size-over-age distribution of firms. The utility cost of default ξ_d is 0.15 and the retention rate of assets upon default χ is 25 percent. These parameters are important for pinning down default rates. We set the shock intensity for the removal of the default flag λ_d to 0.17 which implies an average duration with default in credit history of 6 years. This is to capture the following features of the US bankruptcy system. Households that file for Chapter 13 bankruptcy enter into repayment plans that last for 3–5 years and those that file for Chapter 7 bankruptcy cannot file again for 6 years.

Calibration results Table 3 shows that overall, despite being over-identified with 22 parameters and 27 moments, our model matches the targeted moments well. Starting with

¹³To convert model values to dollar amounts, we use average income from the SCF (2010) and rescale the debt threshold b by the ratio of average model income to average income in the SCF.

¹⁴These moments are computed after excluding entrepreneurs whose employees exceed 500. To convert model labor units to employee counts, we multiply each entrepreneur’s labor by $\frac{500}{l_L}$.

¹⁵Actual maturity is about 6 years as shown in the next section.

Table 3: Calibration results

Moments	Data	Model	Source
Capital-Output ratio	2.4	2.1	BEA
Labor share (wn/y)	0.6	0.6	BLS
Tax-Output ratio	0.16	0.16	CBO
Exit rate	8.5%	9.1%	BDS
Bankruptcy rate	3.0%	1.8%	Dun & Bradstreet
Corporate profit-Output ratio	0.07	0.05	BEA
Aggregate leverage	35.1%	32.3%	Flow of Funds
Share of small firms without debt	30%	33.2%	SBCS
Share of small firms with \$1-1,000,000 debt	65.0%	66.4%	SBCS
Leverage of firms in 0-90th asset percentile	0.35	0.33	Crouzet and Mehrotra (2020)
Leverage of firms in 90-99th asset percentile	0.29	0.35	Crouzet and Mehrotra (2020)
Leverage of firms in 99-99.5th asset percentile	0.30	0.27	Crouzet and Mehrotra (2020)
Share of earnings in top 20%	65.0%	52.6%	SCF
Share of earnings in bottom 60%	14.4%	27.0%	SCF
Share of entrepreneurs in top 20% of wealth distribution	26.0%	26.2%	Cagetti and De Nardi (2006)
Entrepreneurship rate	11.1%	10.4%	Cagetti and De Nardi (2006)
Share of firms that hire 1-19	88.3%	94.0%	BDS
Share of firms that hire 20-99	9.8%	4.3%	BDS
Share of firms that hire 100-499	1.6%	0.6%	BDS
Share of firms that hire 500+	0.4%	1.1%	BDS
Employment share of firms that hire 1-19	19.6%	26.5%	BDS
Employment share of firms that hire 20-99	17.8%	7.5%	BDS
Employment share of firms that hire 100-499	14.1%	14.2%	BDS
Employment share of firms that hire 500+	48.5%	51.9%	BDS
Relative size of entrant	27.5%	18.4%	BDS
Relative size of age 0-5 firms	37.3%	26.3%	BDS
Relative size of age 6-10 firms	54.5%	62.3%	BDS

Note: In the SBCS, all firms have 500 or less employees. Using only data from firms with employees, the SBCS moments are the means between 2016 and 2019. Crouzet and Mehrotra (2020) use US Census Bureau's Quarterly Financial Report (QFR) data to create quarterly panel of US manufacturing firms from 1977 to 2014. They categorize firms by book assets and we group firms by entrepreneurs' assets. The BDS data moments are the means over 1984 - 2015 and the BLS data moment is the mean from 1947 to 2016. The tax-output ratio is averaged over 2000-2014. Aggregate leverage is aggregate debt to assets of non-farm, non-financial businesses from the flow of funds (1954-2006). We use the 2010 wave of SCF, and the definition of entrepreneur in Cagetti and De Nardi (2006) is 'self-employed'.

Table 4: Average sales by size group

	0-90th	90-99th	99-99.5th
Data	1.0	15.7	150.9
Model	1.0	18.0	158.3

Note: The data correspond to the moments reported in [Crouzet and Mehrotra \(2020\)](#), which are computed from the US Census Bureau’s Quarterly Financial Report (QFR) using a quarterly panel of US manufacturing firms over the period 1977–2014. While [Crouzet and Mehrotra \(2020\)](#) categorize firms by book assets, we instead group them by entrepreneurs’ assets. All values are normalized by the mean sales of firms in the 0–90th percentile.

Table 5: Additional leverage moments

Moments	Data	Model	Source
Autocorr. of leverage	0.31	0.50	Kauffman Survey
Cor(leverage, revenue per worker)	-0.05	-0.03	Kauffman Survey
Autocorr. of leverage	0.69	0.80	QSFS
Cor(leverage, revenue per worker)	-0.03	-0.01	QSFS

Note: Both the Kauffman Firm Survey (KFS) and QSFS moments are computed after removing year-industry effects at the two-digit NAICS level. To compare model moments with the KFS moments, we aggregate the simulated series to an annual frequency and restrict the sample to firms younger than eight years. For the QSFS moments, we aggregate the simulated series to a quarterly frequency.

firm borrowing, the data shows average leverage varies little across firm size categories, hovering around 30% [Crouzet and Mehrotra \(2020\)](#). Our calibration captures both the level and size independence of leverage. We also do well in reproducing the share of small firms with no debt, and the share with less than one million dollars of debt. Next we come reasonably close to aggregate nonfinancial business leverage and the bankruptcy rate.

The earnings distribution includes workers’ and entrepreneurs’ earnings. The model is able to capture the concentration of earnings in the fifth quintile of the distribution (Table 2) as well as the share of entrepreneurs at the top of the wealth distribution. In the fourth panel of table 3 we see that our model also comes close in reproducing the firm size (employment) distribution. Looking at the 5th panel, we understate the size of entrants but overstate the size of firms aged 6 - 10 years. Hence firm growth in the model is somewhat faster than in the data.

The model also reproduces moments that were not targeted in the calibration. Table 4 reports average sales across firms, grouped by the size of their assets, normalized by the mean sales of firms in the bottom 90th percentile. In the data, firms in the 90th–99th percentile had roughly 16 times more sales than firms below the 90th percentile, while firms in the 99th–99.5th percentile had average sales that were 151 times more. Our model generates size ratios of 18.0 and 158.3, respectively. We match the high concentration seen in the data

despite it not being a calibration target. This suggests that our calibration of productivity and financial frictions has captured a sizable amount of the firm heterogeneity in the data.

The model further matches untargeted moments reflecting the dynamics of firm leverage. Table 5 reports the autocorrelation of leverage and its correlation with revenue per worker, computed from both the Kauffman Firm Survey (KFS) and the QSFS. Leverage is moderately persistent in the KFS sample of young firms, with an autocorrelation of 0.31, and more persistent in the broader QSFS sample, at 0.69; the model generates values of 0.50 and 0.80, respectively. While leverage is more highly correlated, our model reflects the empirical finding that it is less persistent for young firms.

Across both datasets, leverage is essentially uncorrelated with revenue per worker (-0.05 and -0.03). Surprisingly, the model closely reproduces this near-zero correlation (-0.03 and -0.01). The ability of the model to match both the persistence and the cross-sectional covariance of leverage with labor productivity, without targeting these moments, implies that we have captured critical elements of the link between financial frictions, firm credit, and firm dynamics.

Computation We solve the model in continuous time using finite difference methods, following Achdou et al. (2017). Entrepreneurs’ discrete stopping choices—default, refinancing, entry, and exit—are cast as linear complementarity problems. Using the Broyden algorithm, we find the prices that clear the capital and labor markets, first in the stationary equilibrium and then along perfect-foresight transition paths. Appendix A gives an overview of the numerical method.

4 Steady state analysis

Two features of our model, entrepreneurs with diminishing marginal utility and illiquid long-term debt, shape the distribution of producers and, through it, the allocation of resources. We now compare the calibrated benchmark model with alternative versions in steady state, each of which weakens or removes one of these features. These alternatives involve (i) shorter-duration loans, (ii) weaker consumption-smoothing motives and (iii) no long-term debt. We then turn to the cross-sectional dispersion of entrepreneurs’ stochastic discount factors, which indicates how poorly idiosyncratic risk is shared across entrepreneurs in each variant. In Section 5 we show that cyclical changes in this uninsurable risk are important determinants of the model’s response to aggregate shocks.

There is considerable dispersion in assets and debt across entrepreneurs in the steady

state.¹⁶ A result of this dispersion is the misallocation of resources. The collateral constraint keeps entrepreneurs with too few assets from operating at their efficient scale; hence, the distribution of entrepreneurs over assets affects aggregate output. At the same time, the distribution of debt determines disposable income and thus savings and the future distribution of assets.

Diminishing marginal utility and illiquid debt, together, set our environment apart from related models. Relative to models of risk-neutral firms such as [Khan and Thomas \(2013\)](#), our entrepreneurs must consume and may dissave even when constrained, whereas the firms in [Khan and Thomas \(2013\)](#) delay dividends to accumulate capital when constrained. Relative to models of entrepreneurs such as [Buera et al. \(2011\)](#) and [Buera and Shin \(2013\)](#), long-term debt gives producers in our model a way to dampen the effects of collateral constraints.

We examine the three alternatives. First, we create shorter-duration loans by lowering the refinancing cost parameter ξ_b from 0.03 to 0.¹⁷ While loan duration could be infinite absent refinancing, prepayment or default, in practice entrepreneurs adjust or terminate their contracts as shocks change their desired portfolio. Second, we weaken the consumption-smoothing motive by reducing the coefficient of relative risk aversion from 3 to 2. In this *less consumption smoothing* case, entrepreneurs are more willing to let consumption vary, so they enter production sooner and take on more leverage. Third, we remove long-term debt entirely. Entrepreneurs can then no longer borrow to build up their assets; there is no default, and their stopping choices collapse to just entry and exit. This version, close to [Buera et al. \(2011\)](#) and [Buera and Shin \(2013\)](#), isolates the role of debt.

4.1 Role of long-term debt and risk-averse producers

Table 6 compares the benchmark with the three alternative models. The model without debt and the model with shorter duration help illustrate the impact of illiquid, long-term debt. Removing debt sharply worsens the misallocation caused by the collateral constraint. When entrepreneurs cannot borrow, they need higher savings to pay entry costs. Once active, they also need to rely on assets to increase capital spending. As a result, only 27% of entrepreneurs engage in production, output is 20% lower, and aggregate total factor productivity is 13% lower than in the benchmark case. The presence of risk-neutral firms mutes these aggregate effects; restricting attention to entrepreneurs alone, output falls by 83% and TFP by 26%.

While debt helps dampen the allocative distortions arising from collateral constraints, its illiquidity increases misallocation. In the shorter-duration case, reducing the proportional

¹⁶Figure 16 in Online Appendix E shows the steady-state distribution.

¹⁷The refinancing cost is $\xi_b|b'| + \xi_{b_0}$, so refinancing is not free even when $\xi_b = 0$.

Table 6: Role of long-term debt

	Benchmark	Shorter duration	Less cons. smoothing	No debt
Output	1.00	1.02	1.02	0.80
Entrepreneur's output	1.00	1.06	1.37	0.17
Interest rate	1.00	0.94	1.29	2.32
Wage	1.00	1.01	1.03	0.80
TFP	1.00	1.00	1.08	0.87
Entrepreneur's TFP	1.00	0.99	1.06	0.74
Active entrepreneurs (%)	34.51	34.60	59.18	26.78
Aggregate leverage	0.32	0.40	0.37	-
Loan duration (quarters)	23.3	9.2	14.3	-

Note: ‘Shorter duration’ is a version of the model with a smaller refinancing cost. ‘Less consumption smoothing’ is a version with a smaller curvature on the entrepreneur’s utility. ‘Without debt’ is a version without the long-term debt b and stopping options involving the long-term debt. Output, interest rate, wage, and TFP are normalized to the benchmark model value. The loan duration is average of time computed from simulating a panel of 20,000 entrepreneurs for 1,200 quarters.

costs of debt adjustment leads to more frequent refinancing and prepayment; the effective duration of the loan falls to 9.2 quarters from 23.3 quarters. As entrepreneurs find it easier to adjust their financial positions, their output rises by 6% and total output by 2%.

The less-consumption-smoothing model isolates the role of diminishing marginal utility. Lowering the curvature of utility makes entrepreneurs more tolerant of consumption swings, weakening the precautionary motive and strengthening the incentive to enter and to borrow. The share of active entrepreneurs nearly doubles, from 34.5% to 59.2%, and those who produce take on more leverage: aggregate leverage rises from 0.32 to 0.37, and the effective loan duration shortens from 23.3 to 14.3 quarters as entrepreneurs adjust their debt more aggressively. With more producers using more debt to operate closer to the scale of unconstrained firms, entrepreneurs’ output is 37% higher and TFP 8% higher than in the benchmark model. The rise in TFP raises the interest rate by 29% and wages by 3%; these higher input prices crowd out risk-neutral firms, so aggregate output rises by only 2%. Greater tolerance of consumption risk thus lowers steady-state misallocation.

4.2 Dispersion in the stochastic discount factor

The consequences of long-term debt and the collateral constraint for consumption risk are vivid in the cross-section of entrepreneurs’ stochastic discount factors. Each household values an extra unit of resources in a future state at its marginal rate of substitution, so its

one-period stochastic discount factor (SDF) is

$$m_{i,j} = e^{-\rho \Delta t} \frac{u'(c'_{i,j})}{u'(c_i)}, \quad (7)$$

where $\Delta t = 0.25$ is one quarter, ρ is the discount rate, c_i is consumption in state i , and $c'_{i,j}$ is consumption in a state reachable next quarter. Recall that when idiosyncratic shocks are fully insured, all entrepreneurs share a common SDF; with incomplete markets, SDFs differ across individuals. Therefore, the cross-sectional dispersion of $m_{i,j}$ measures how unequally entrepreneurs are exposed to consumption risk. A high $m_{i,j}$ identifies an entrepreneur who may experience a sharp fall in consumption, and any dispersion implies consumption risk. We find below that this dispersion rises with risky debt. In the next section, we show that this leads entrepreneurs to deleverage in a downturn.

Table 7 reports the first four moments of the discount factors implied by the SDF distribution for the benchmark version of the model and the version without debt. When computing moments, we weight each realization by the product of the one-quarter transition probability and the stationary mass of active entrepreneurs at the associated current state. A handful of transitions lead to very low consumption states with $m_{i,j}$ very large. We therefore drop the top one percent of realizations before computing moments. Figure 1 plots the corresponding histograms.

Table 7: Cross-sectional moments of entrepreneurs' discount factors

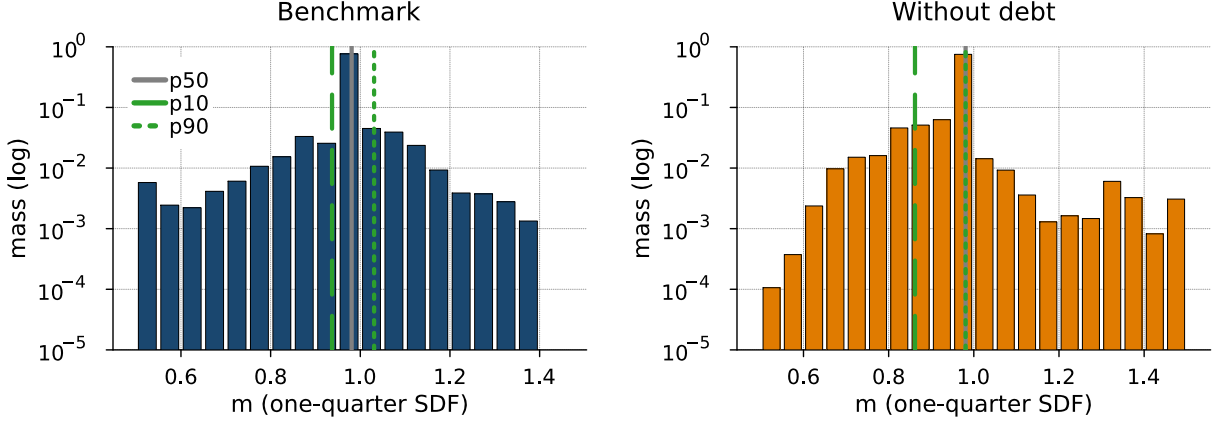
	Mean	Std. dev.	Kelley skew.	Excess kurtosis
Benchmark	0.977	0.089	0.07	30.7
Without debt	0.964	0.100	-1.00	71.8

Note: The one-quarter SDF of entrepreneur in state i is $m_{i,j} = e^{-\rho \Delta t} u'(c'_{i,j})/u'(c_i)$ with $\Delta t = 0.25$ (Equation 7). All statistics are computed over the entrepreneurs in the steady-state distribution, weighting each realization m_{ij} by the joint mass $g_i p_{ij}$ (stationary mass on the starting state times the one-quarter forward transition probability). The mean, standard deviation, and excess kurtosis are the mass-weighted moments. The Kelley skewness is the decile-based measure $(Q_{90} + Q_{10} - 2Q_{50})/(Q_{90} - Q_{10})$, where Q_p is the mass-weighted p -th percentile. The top one percent of realizations by mass are dropped before computing all statistics.

Table 7 and Figure 1 reveal a sharp contrast between the versions of the model with and without debt. In the benchmark, the mean discount factor is about 0.98, while without debt it falls to 0.964.¹⁸ When entrepreneurs hold debt, a larger fraction risks a substantial fall in consumption growth if there is a negative productivity shock between state i and state

¹⁸The results from the model with shorter debt duration and a weaker consumption-smoothing motive are similar to those from the benchmark model.

Figure 1: Distribution of the entrepreneur stochastic discount factor



Note: Each panel plots the mass-weighted histogram of the one-quarter SDF, $m_{i,j} = e^{-\rho \Delta t} u'(c'_{i,j})/u'(c_i)$, across entrepreneurs in the steady state of each model version, after dropping the top one percent of realizations by mass. The vertical axis is on a log scale to reveal the tail bins. The vertical lines mark the mass-weighted 10th, 50th, and 90th percentiles of the distribution (p50 solid, p10 dashed, p90 dotted).

j. An entrepreneur's disposable income is $\pi_i - (\theta + r(\Omega))b + ra - \text{tax}$, so when a negative idiosyncratic shock lowers profit from π_i to π_j , the fixed debt-service payment $(\theta + r(\Omega))b$ absorbs none of the loss. Consumption falls by more than it would in the absence of debt. Consequently, the mass of states with high SDF is larger in the model with debt.

The Kelley skewness reinforces this finding. Without debt, a large share of entrepreneurs sit at $m \approx e^{-\rho \Delta t}$ —the value associated with flat consumption—so the 50th and 90th percentiles of the SDF coincide, most of the mass lies below one, and the Kelley skewness equals -1 . With long-term debt, the debt-service channel stretches out a right tail: the 90th percentile moves above the median, and the Kelley skewness turns positive (0.07). Figure 1 shows this. The vertical lines mark the 10th, 50th, and 90th percentiles: without debt, the median and 90th percentile bunch together below one, whereas with debt they spread apart and the 90th is above the median, on a distribution that carries visible mass above one.¹⁹

A positively skewed SDF in the benchmark means some entrepreneurs bear high risk. Existing debt obligations mean that those already with low consumption would experience a sharp rise in marginal utility from a negative income shock. These are entrepreneurs with the strongest precautionary motive to cut debt when a recession raises their consumption risk from further income drops. We will see the aggregate impact of their actions next.

¹⁹The 95th percentile of the stochastic discount factor is 1.092 in the benchmark and 0.981 in the economy without debt; the 99th percentiles are 1.219 and 1.329, respectively.

5 Results

Recent approaches to understanding the business cycle implications of costly external finance have emphasized the role of firm heterogeneity.²⁰ Compared to such models, our setting has long-term, illiquid debt held by entrepreneurs. We show that the principal payments associated with long-lived debt coupled with risk-averse borrowers create nonlinear individual responses to aggregate shocks.

In section 5.1 we show that these nonlinear individual decision rules lead to shifts in the distribution of entrepreneurs over assets and debt. Movements in this distribution drive changes in misallocation over the business cycle. This leads to aggregate nonlinearities. In recessions, the distribution of production shifts away from highly productive but low net worth entrepreneurs. However, expansions see little reallocation of production. Changes in the allocation of capital and labour across firms, in a model where firms produce using decreasing returns to scale technologies, affect aggregate total factor productivity. In a recession, shifts in the distribution of firms, reduce TFP. This provides an aggregate measure of increased misallocation.

In section 5.2 we relate these distributional movements directly to a firm-level measure of misallocation, dispersion in the marginal revenue product of capital. We show that counter-cyclical increases in this dispersion are associated with the reductions in TFP found in the previous section. Section 5.3 examines the role of time-varying misallocation in the Great Recession. We find that a fall in the value of business assets led to a protracted rise in misallocation that helps explain the slow recovery from the recession.

5.1 Misallocation over the business cycle

We begin our analysis of the model over the business cycle by studying modest, one percent exogenous shocks to TFP. To study a recovery preceded by a recession, we assume that the exogenous component of TFP falls by 1% and remains at this level for several quarters. Afterwards, we allow a fairly rapid return to its mean.²¹ ²² Below, we show that endogenous responses in aggregate TFP amplify the exogenous shock by 60 percent in a recession.

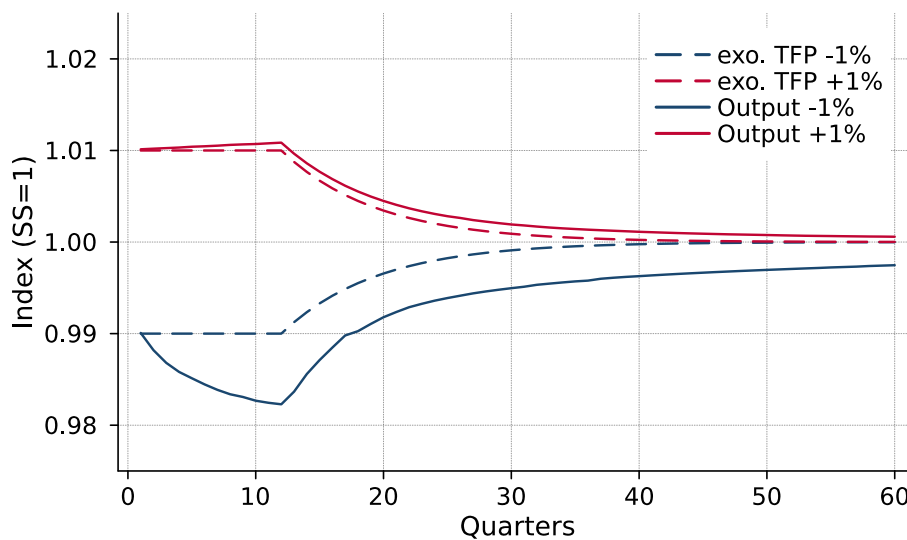
²⁰Khan and Thomas (2013) argue that the aggregate debt-to-asset ratio in the flow of funds implies that a representative firm would not need external financing of investment. Models with a distribution of firms can be consistent with this aggregate data while simultaneously allowing a nontrivial role for debt among a subset of firms.

²¹Specifically, we study positive and negative shocks to aggregate TFP that last for 12 quarters, then return to their mean with a persistence of 0.5. While the shocks have time-varying persistence, their overall half-life is comparable to that of shocks with constant persistence of about 0.9.

²²Unanticipated shocks may generate aggregate profits or losses in the financial sector; we assume that these are absorbed by the government. This allows us to avoid introducing a market for shares in financial intermediaries.

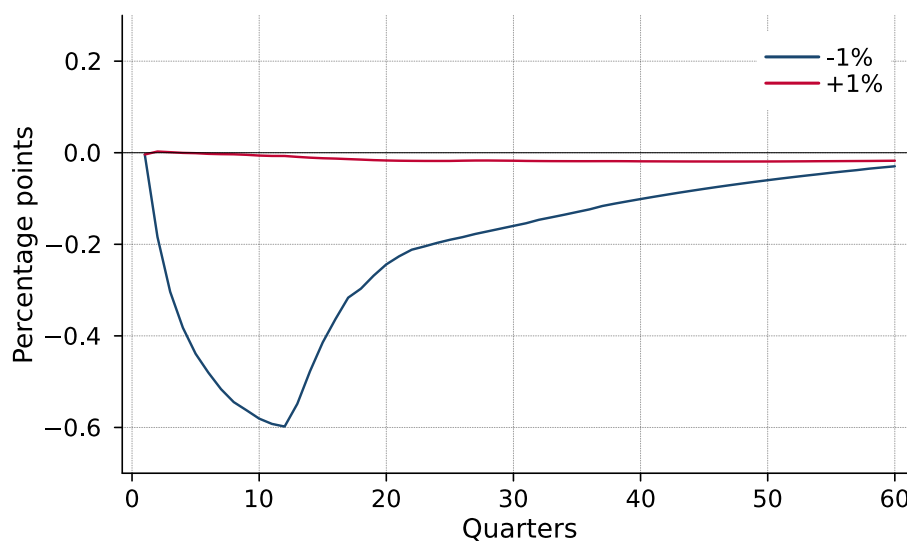
However, there is negligible endogenous response in an expansion.

Figure 2: GDP and exogenous TFP in expansions and recessions



Note: Output is normalized by its steady state level.

Figure 3: TFP net of exogenous component



Note: Series shown are deviations from their steady state values.

Output and endogenous productivity Figure 2 plots exogenous TFP and output paths.

The dashed lines show the exogenous change in aggregate TFP—red in the expansion, blue in the recession—and the solid lines show the corresponding paths of GDP. The trough of GDP in the recession is at date 12 when it falls -1.78% below its long-run average. The peak of GDP in the expansion is at the same date, but here its rise is 1.08%. Thus GDP falls by 0.7 percentage points more, following a -1% shock, than it rises from a positive shock of the same magnitude. This implies significant propagation of the exogenous shock in the recession, while there’s little amplification of the shock over the expansion. This asymmetry is absent in a version of the model without debt, as shown in Figure 17 of Appendix E.

The sharper contraction in the recession is entirely due to a rise in misallocation across firms. Figure 3 plots the difference between TFP and its exogenous component, again blue in the recession and red in the expansion.²³ In the recession, TFP falls well below the exogenous shock: by the start of the recovery it has fallen 0.6 percentage points more than the exogenous component, so rising misallocation propagates the fall in TFP by 60 percent. When viewed in terms of GDP, the recession is 65 percent deeper than the corresponding expansion. Indeed, TFP actually rises by less than the exogenous shock in the expansion, mildly dampening its effect on GDP.

Output decomposition These movements in the endogenous component of TFP arise from the redistribution of capital and labor across active entrepreneurs, and this redistribution operates through two channels. First, at each individual state (z, a, b) , entrepreneurs change their choices of capital k and labor l . Second, the measure of entrepreneurs across states shifts. To separate the two channels, let i index the distribution of producers over (z_i, a_i, b_i) , and let g_{i*} denote the steady state density and g_{it} the time- t density. An active entrepreneur of type i produces $A_t z_i k_{it}^\alpha l_{it}^\nu$. In any period, total entrepreneurial output y_t can be decomposed as

$$\begin{aligned} y_t &= A_t \int z_i k_{it}^\alpha l_{it}^\nu g_{it} di = A_t \int z_i k_{it}^\alpha l_{it}^\nu (g_{i*} + \Delta g_{it}) di \\ &= A_t \int z_i k_{it}^\alpha l_{it}^\nu g_{i*} di + A_t \int z_i k_{it}^\alpha l_{it}^\nu \Delta g_{it} di. \end{aligned} \tag{8}$$

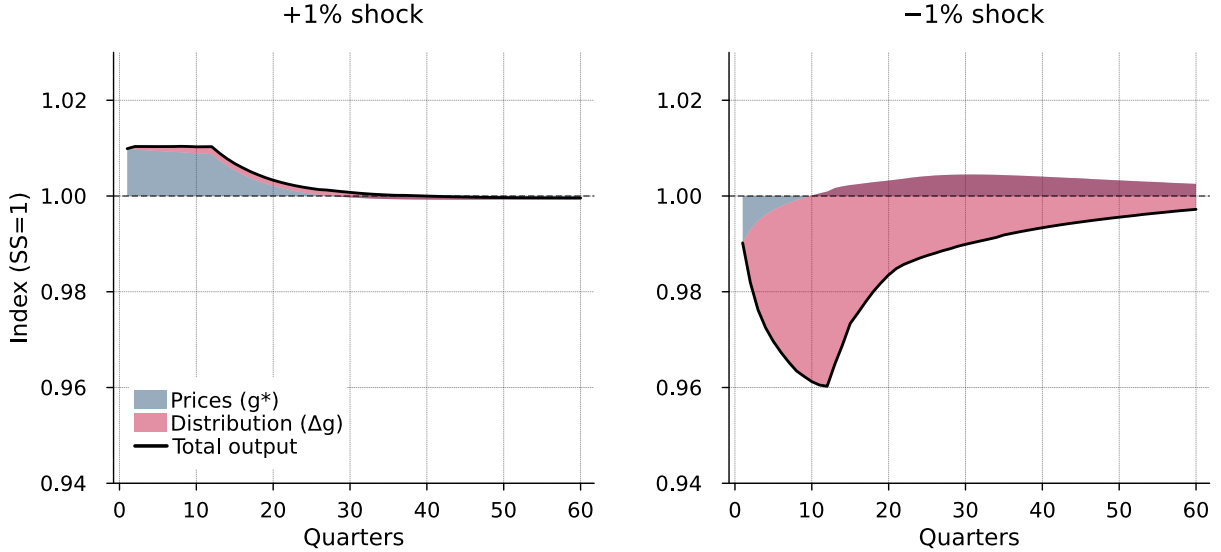
where $\Delta g_{it} = g_{it} - g_{i*}$.

Figure 4 shows this output decomposition. The left panel plots the +1% expansion and the right panel the -1% recession; in each, the black line is entrepreneurs’ output and the two shaded bands are the two terms of Equation 8, measured from the steady state and

²³TFP is $\frac{Y}{K^\alpha L^\nu}$, where Y is GDP and K and L are the aggregate stocks of capital and labor.

summing to the total. Entrepreneurs' production falls 4% in the recession—far more than the 1.78% fall in aggregate output in Figure 2—because production shifts toward risk-neutral firms, which benefit from the recessionary decline in real wages and rental rates.

Figure 4: Decomposing changes in output of entrepreneurs



Note: Each panel decomposes entrepreneurs' output, normalized by its steady-state level, over the $\pm 1\%$ TFP shock: the left panel shows the $+1\%$ expansion and the right panel the -1% recession. In each, the black line is entrepreneurs' total output, and the two shaded bands are the two terms of Equation 8, measured from the steady state and summing to it. The *Prices* band is the first term, which captures price-driven changes in decision rules holding the distribution of entrepreneurs fixed at g_{i*} ; the *Distribution* band is the second term, which captures the effect of the shift in the distribution Δg_{it} .

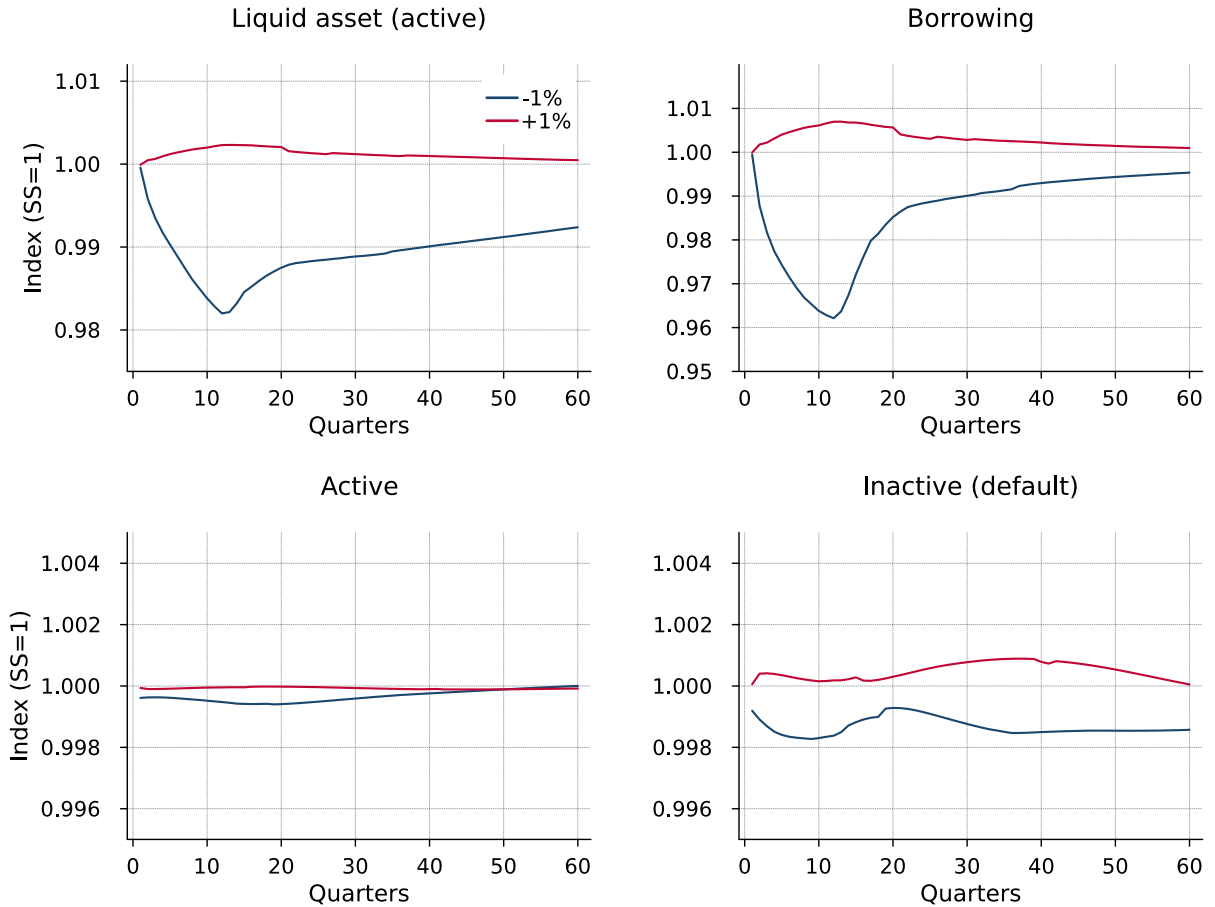
The two shaded bands decompose entrepreneurs' production and sum to the total in each panel. The *Prices* band is the first term of Equation 8: it aggregates actual production at each point, $y_{it} = A_t z_i k_{it}^\alpha l_{it}^\nu$, using the steady-state density g_{i*} , and so shows how output would move from time-varying changes in capital and labor—at each level of productivity, debt, and assets—if the distribution of entrepreneurs had not moved. The *Distribution* band is the second term: it weights the change in the distribution, $\Delta g_{it} = g_{it} - g_{i*}$, by equilibrium production, and so captures the effect of shifts in the number and composition of entrepreneurs.

In the recession (right panel), the initial fall in entrepreneurs' output is driven by changing decision rules, the *Prices* band. Eventually falling wages and interest rates imply that these decision rules, aggregated with the steady-state distribution, would raise production, so the *Prices* band turns positive. The expansion (left panel) works almost entirely through this channel: the response of decision rules accounts for nearly all of the rise in output, with almost no contribution from the *Distribution* band. The recession is different. There the

Distribution band shows that the rising misallocation that lowers TFP in Figure 3 comes entirely from changes in the distribution of entrepreneurs. Not only does this distributional effect eventually explain the entire recession, it is persistent. Overall, it drives a deeper recession than expansion and a more gradual recovery.

Precautionary deleveraging The shift in the distribution of entrepreneurs increasing the misallocation of resources in a recession may arise through a rise in exit or from continuing entrepreneurs experiencing changes in their assets and debt. The lower panels of Figure 5 show the share of active entrepreneurs together with the share of inactive entrepreneurs with a default in their credit history; neither changes much following a one-percent shock to TFP.

Figure 5: Assets, debt and entrepreneurship



Note: Blue lines correspond to the recession while red lines represent the expansion. All series are normalized by their respective steady state levels.

In the top panels of Figure 5, we see little change in assets and debt in the expansion. Over the first 13 quarters, assets rise 11 basis points above their steady state level and debt

risers 70 basis points. The recession is sharper. Assets fall by almost 2%, and debt falls even more, 3.8%, seen in the upper right panel. This reduction in debt is financed by assets. Because a significant share of entrepreneurs (66% in the steady state) are subject to the collateral constraint, the fall in assets lowers their capital relative to their productivity. This increases misallocation and reduces TFP.

Figure 6: Changes in leverage across entrepreneurs

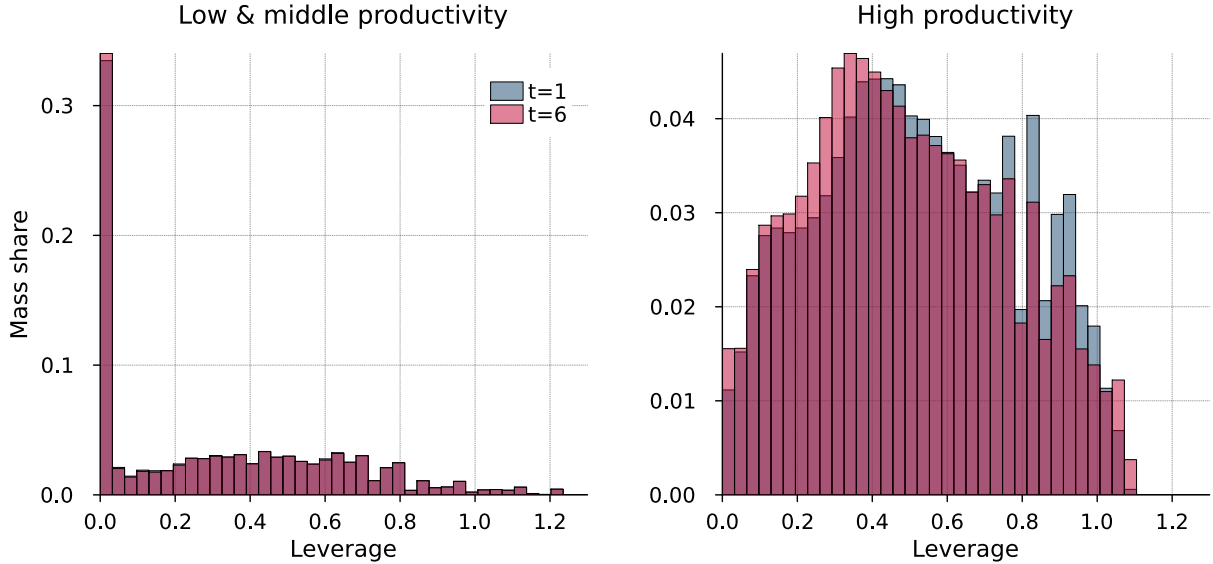


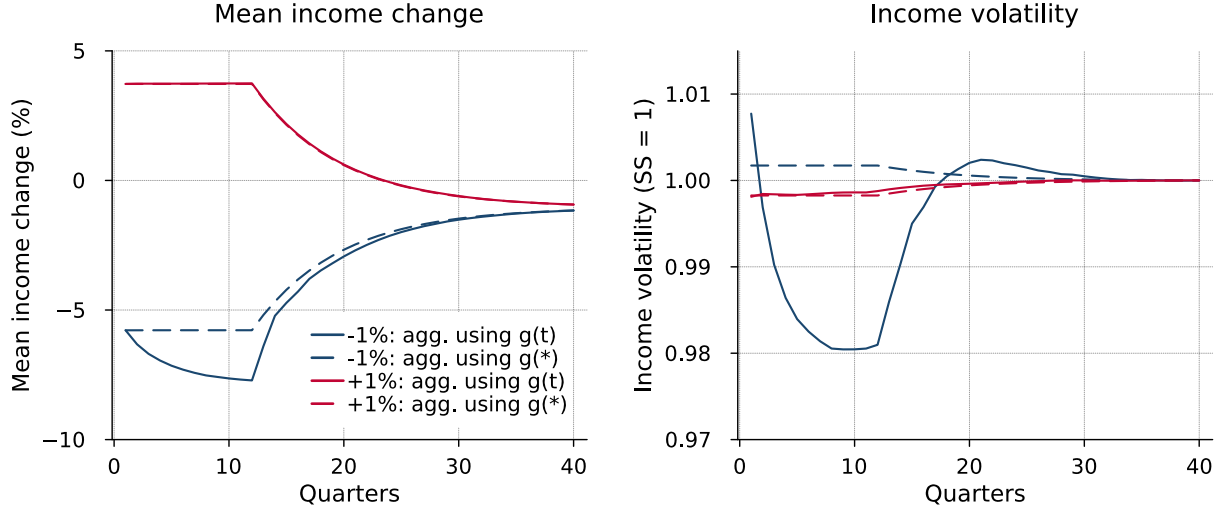
Figure 6 shows the cross-sectional distribution of leverage across entrepreneurs, at the pre-shock steady state and six quarters into the recession, split by productivity.²⁴ Among low- and middle-productivity firms (left panel), the distribution shows little change. Among the most productive firms (right panel), the distribution shifts to the left indicating deleveraging. The associated movements in the assets and debt of high-productivity firms are responsible for a rise in misallocation.

To measure that misallocation, we use the coefficient of variation of capital across firms. As capital is rented in spot markets, this would be 0 absent collateral constraints. There is no notable cyclical in this measure for low- and mid-productivity firms, they remain at their steady state levels of 0.895 and 0.933, respectively.

The steady state coefficient of variation of capital for high productivity firms is 0.56. It remains unchanged in an expansion, as suggested by the flat response of average leverage. However, it is only among such firms that it rises in a recession, to 0.62 (Figure 19 in Appendix). Misallocation therefore increases among the most productive firms, those that

²⁴The full path of average leverage for each productivity type, over both expansion and recession, is shown in Appendix Figure 18.

Figure 7: Income growth and volatility of high-productivity entrepreneurs



Note: The left panel shows the expected income change of the highest-productivity entrepreneurs; the right panel shows their income volatility, indexed to its steady state value. Blue lines correspond to the recession and red lines to the expansion. Solid lines aggregate using the actual distribution at each date, $g(t)$, while dashed lines use the steady state distribution, $g(*)$.

deleverage.

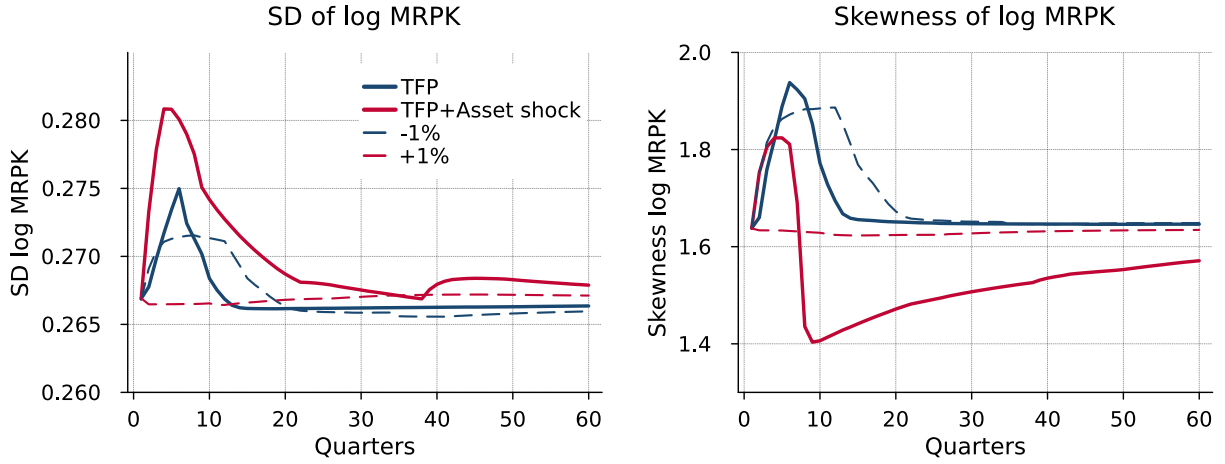
In summary, the recessionary rise in aggregate misallocation, the gap between aggregate TFP and its exogenous component in Figure 3, reflects increased dispersion in capital across high-productivity entrepreneurs. The dispersion arises as they draw down their assets to reduce debt and deleverage (the right panel of Figure 6), and the resulting changes in their holdings of assets and debt are the source of the distributional effect in Figure 4. As the most productive entrepreneurs shed assets in the recession, they rent less capital and their share of output falls relative to less productive firms.

We explore the causes behind high-productivity entrepreneurs deleveraging. Figure 7 plots their expected income growth and income volatility. In each panel we aggregate individual outcomes two ways. We use the actual distribution at each date, $g(t)$, and the steady state distribution, $g(*)$. The gap between the two isolates the effect of the shifting distribution over assets and debt.

Income growth rises in expansions and falls in recession. In the latter, the decision to deleverage by high-productivity firms shifts their distribution in a way that deepens the fall in their income. The right panel suggests why they do so. A recession raises income volatility which deleveraging subsequently reduces. Together, the two panels show that the most productive firms accept lower income growth in order to reduce their income risk.

Precautionary deleveraging results from the combination of fixed payments on long-term

Figure 8: Dispersion and skewness of log MRPK over the business cycle



Note: Each panel plots a moment of the cross-sectional distribution of log MRPK across active entrepreneurs along the perfect-foresight transition. MRPK is $\alpha A z k^{\alpha-1} l^{\nu}$.

debt and risk-averse borrowers. To isolate the role of each element, in Appendix B we solve the model's equilibrium path under three variations, a weaker consumption-smoothing motive, shorter debt duration, and lower debt payments. The exercise shows that output falls less when the consumption-smoothing motive is weaker, while shorter duration and lower payment burdens mainly speed up the recovery.

5.2 Countercyclical dispersion

We now compare the degree of misallocation over the business cycle in our model against the empirical evidence. Led by the influential works of Restuccia and Rogerson (2008) and Hsieh and Klenow (2009), a large literature studies how the misallocation of resources across firms lowers aggregate total factor productivity. Far less is known, however, about how misallocation moves over the cycle. The study most relevant for us is Alam (2020), who documents that marginal revenue product of capital dispersion is countercyclical across nine European economies, Canada, and the United States. For the United States, he uses Compustat data from 1961 to 2014 to compute the cyclical component of the industry-weighted standard deviation of log MRPK, and finds in his Figure 2(a) that this measure rose 2.4 percentage points between 2007 and 2009. As detrended GDP fell 4.15 percent over the same period, the rise in dispersion amounts to 58 percent of the fall in GDP.

In the left panel of Figure 8 we see that dispersion in the returns to capital is, indeed, countercyclical. The panel plots the standard deviation of the log of entrepreneurs' marginal

revenue products of capital along four impulse responses. The first two responses are the +1 and −1 percent TFP shocks of Section 5.1; the remaining two, labeled *TFP* and *TFP+Asset*, are the Great Recession shocks studied in the next section. Both are calibrated to reproduce the fall in TFP over that episode. The *TFP+Asset* case adds a shock to entrepreneurs’ assets.²⁵

The rise in dispersion is largest in the *TFP+Asset* case, where the standard deviation of log MRPK increases by about 1.4 percentage points. As we show below, GDP falls roughly 2.7 percent in this case which is a little over half of the 5.1 percent decline in real GDP between 2007Q4 and 2009Q2. We see that our model generates a rise in dispersion equal to 52 percent of the fall in GDP. To sum up, in both model and data, dispersion is countercyclical. Further, for the Great Recession, the rise in dispersion, relative to GDP, predicted by our model is almost the same as the empirical measure in Alam (2020).

While there is some empirical evidence on the cyclicity of dispersion, there is none on skewness. In the model, the average distribution of log marginal revenue product of capital is substantially right-skewed averaging around 1.64. Moreover, this skewness is countercyclical. As more productive but financially constrained firms deleverage, the rise in their marginal revenue products of capital skews the distribution to the right. This change in skewness is substantial, it rises by 0.24 for our Great Recession case with both TFP and Asset shocks. Interestingly, this increase is greater for the case with a larger TFP shock and no asset shock. As the asset shock is of the same proportion across all firms, it shifts the entire distribution to the right. While this involves a rise in variance, a right-ward shift in the left tail, where most firms were financially unconstrained, actually reduces skewness.

5.3 Misallocation and Leverage over the Great Recession

The Great Recession involved extraordinary changes in firms’ balance sheets. Using our model, we examine the aggregate effects of these changes for the aftermath of the recession. Our goal is not to explain the causes of the recession nor its severity; we abstract from credit shocks to collateral constraints or financial intermediaries’ technology for making loans.²⁶ Instead, we focus on how the recovery was affected by the sharp fall in the value of business assets that occurred between 2007Q4 and 2009Q3. In the following, we introduce unanticipated shocks that reduce the value of entrepreneurs’ assets; these coincide with exogenous shocks to total factor productivity.

²⁵As explained below, the asset shock allows us to reproduce the fall in the value of nonfinancial business assets over the Great Recession.

²⁶Credit shocks operating through collateral constraints are studied by Jermann and Quadrini (2012) and Khan and Thomas (2013); shocks to financial intermediaries’ lending technology by Khan et al. (2021).

Our asset shocks reproduce the 15 percent fall in nonfinancial business assets over the first eight quarters of the recession.²⁷ We reproduce it in the model with a series of proportional shocks to active entrepreneurs' assets over these eight quarters, calibrated so that the path of their assets tracks that of nonfinancial business assets and falls by a cumulative 15 percent (left panel of Figure 10). Mechanically, the assets entrepreneurs lose are transferred to the government, so the shocks reallocate the ownership of capital rather than destroy it. The government rents this capital back to firms and rebates the earnings lump-sum to all households, both workers and entrepreneurs. In the absence of investment, its holdings depreciate away over time. The shock entrepreneurs experience is thus purely financial. No capital is lost, and, absent financial frictions, it would leave the allocation of productive resources unchanged.

Total factor productivity fell 2.7% against trend between 2007Q4 and 2009Q1. Given the asset shocks, we match the 2.7% fall in TFP using a -1.36% shock to its exogenous component. We label this model with two shocks $TFP + Asset$. We isolate the role of changes in the value of assets by comparing the two shock model to an alternate single shock case which also matches the empirical fall in TFP, TFP only. Here we need a larger shock of -2.0%. In summary, we compare two models. Both are consistent with the -2.7% change in TFP in the Great Recession. The first model further reproduces the 15% fall in the value of business assets. As a fall in assets increases misallocation thereby lowering TFP endogenously, this model needs a smaller exogenous shock to TFP. Figure 9 plots the exogenous shocks, the resulting paths of TFP, and the gap between them.

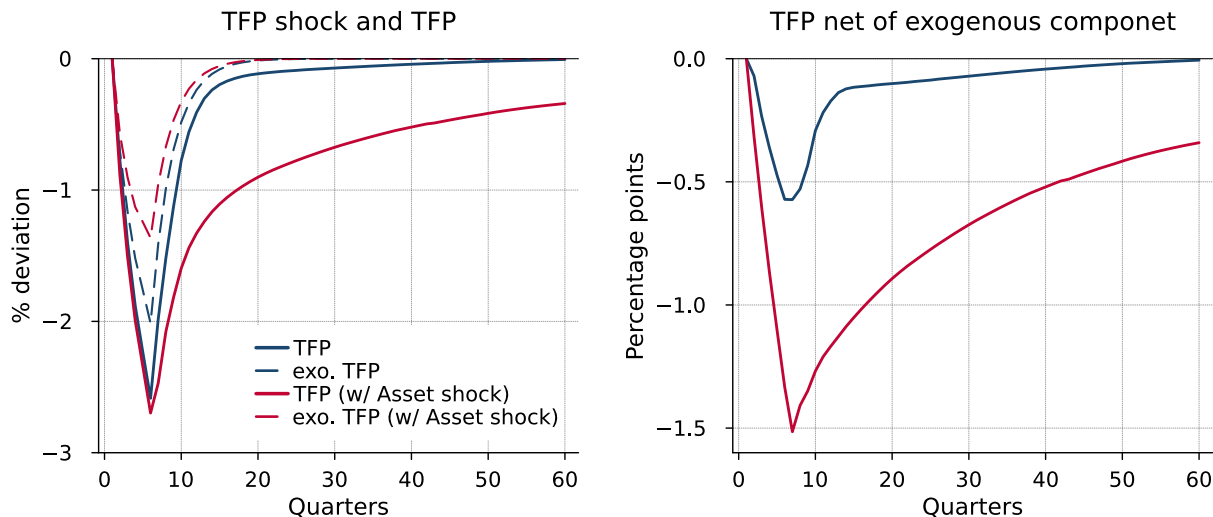
Although we impose shocks to entrepreneurs' assets, they respond by adjusting debt and the path of leverage is endogenous. The right panel of Figure 10 shows that, by matching the observed loss of assets, the $TFP + Asset$ model reproduces almost 60 percent of the rise in leverage observed over the recession.

Given collateral constraints, a fall in the value of a producer's assets lowers the capital she can operate, pushing constrained producers further below their efficient scale of production and raising misallocation. It follows that both the asset and TFP shocks increase misallocation. This is why the gap between TFP and the shock falls further in the $TFP + Asset$ case (Figure 9).

Reproducing the fall in the value of business assets increases the persistence of the recession. Figure 11 shows output and the exogenous TFP shock for both experiments alongside the data. Both experiments produce a similar initial decline, about 2.7% by quarter six, but their recoveries differ substantially. Output recovers much more slowly under the asset

²⁷The decline in nonfinancial business assets is computed from the Financial Accounts of the United States (Z.1), Board of Governors of the Federal Reserve System.

Figure 9: TFP: exogenous shock and endogenous misallocation



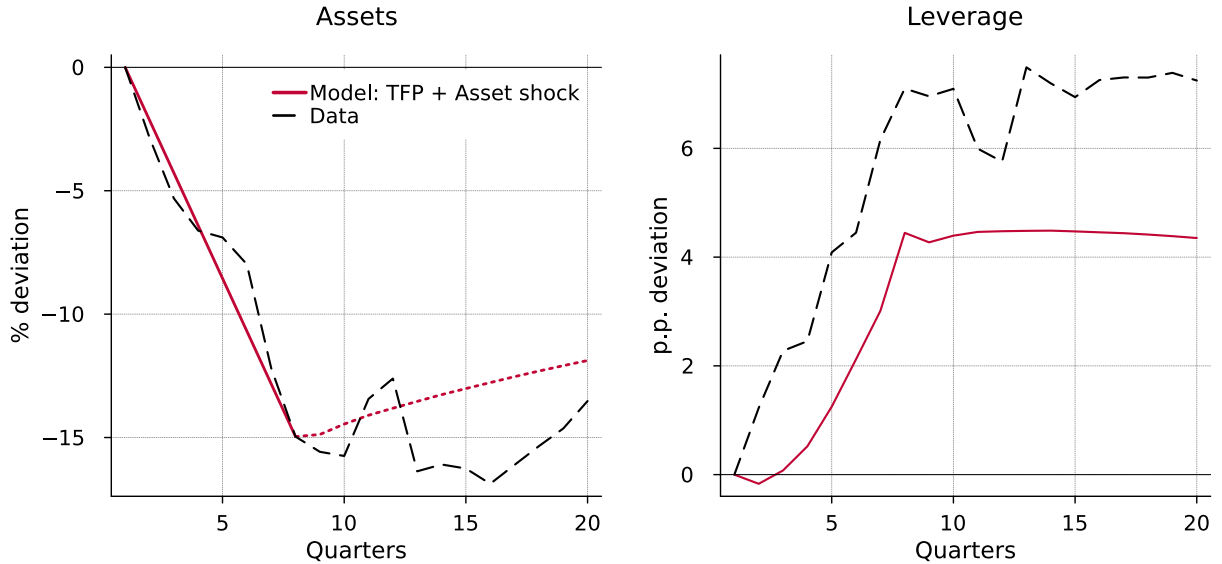
Notes: The figure decomposes the fall in TFP over the Great Recession into its exogenous and endogenous parts, for the two experiments described in the text: *TFP only* (blue) and *TFP + Asset* (red), aligned so that 2007Q4 corresponds to model period 1. Panel (a) plots the exogenous shock to TFP (dashed) alongside the resulting TFP (solid), each as a percentage deviation from the pre-shock steady state. Panel (b) plots the gap between them.

shock: the downturn's half-life increases from roughly 8 quarters to roughly 20, a recovery that is 2.5 times slower. The actual recession was far more persistent—forty quarters after it began, GDP remained more than 2% below trend. The two-shock model captures at least half of that persistence, because the loss of assets sets off a large and long-lived rise in misallocation (left panel of Figure 8). Abstracting from credit shocks, it reproduces roughly half of the actual fall in GDP and, arguably, as much of its persistence; had the model downturn been a percentage point deeper, its recovery would have overlain the data.

Figure 12 decomposes entrepreneurs' output in each experiment into the price and distribution terms of Equation 8. The distribution term is about four times larger in the *TFP + Asset* case, and entrepreneurs' production falls almost twice compared to the TFP shock alone. While GDP falls about 2.7%, output in the entrepreneurial sector falls 10.2%. The response of risk neutral firms to the fall in real wages and interest rates dampens the aggregate fall in GDP. Nonetheless there is a protracted recession.

The source of this persistence is the balance sheet. While the TFP shock decays quickly, the fall in assets does not: the upper-left panel of Figure 13 shows that entrepreneurs' assets remain low long after productivity has recovered. Debt falls by a similar amount in both experiments, but while assets change little in the *TFP only* case, the balance-sheet shock cuts them by 15%. For financially constrained entrepreneurs, the loss of income from lower

Figure 10: Assets and Leverage: Model and Data



Notes: The figure compares the model's *TFP + Asset* experiment with U.S. nonfinancial business-sector data over the Great Recession, aligning 2007Q4 with model period 1. The left panel plots real (CPI-deflated) assets as percentage deviations from their 2007Q4 level; the model line is solid over the first eight quarters, during which the asset shock is imposed, and dotted thereafter. The right panel plots leverage as percentage-point deviations from its 2007Q4 level. Sources: Federal Reserve Financial Accounts of the United States (Z.1); assets are deflated by the consumer price index (CPIAUCSL).

productivity is then exacerbated by a loss of capital. They reduce savings in an effort to dampen the fall in their consumption relative to income. As a result, assets recover slowly.

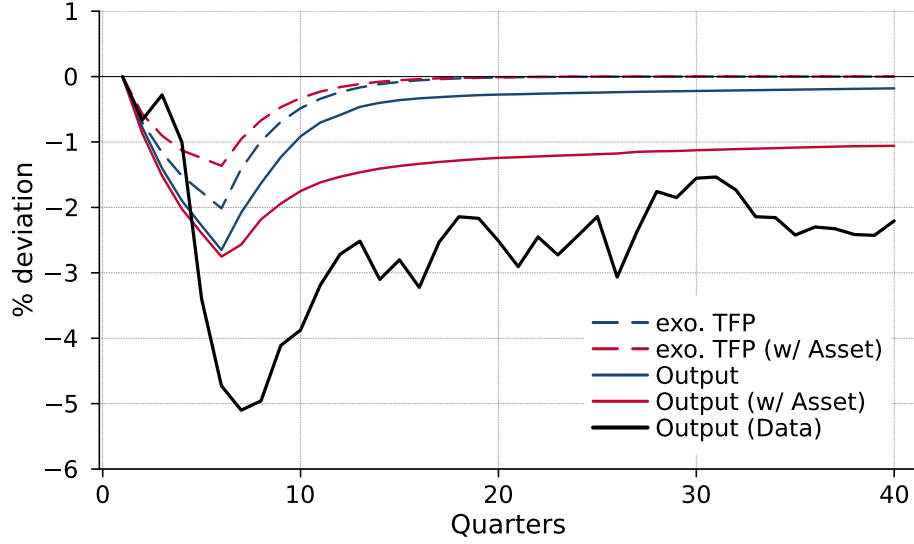
Our model recession explains much of the loss of business over the Great Recession. Whereas the small shocks of Section 5.1 left entry and exit almost unchanged, the lower-left panel of Figure 13 shows the number of producers falling 2.75% in the *TFP + Asset* case. By itself, this is half of the actual decline in businesses between 2007 and 2010.²⁸ However, recall the model explains approximately half the loss in GDP. The *TFP + Asset* case is thus consistent with the change in the number of businesses relative to GDP over the Great Recession.

6 Evidence from firm balance sheets

Aggregate dynamics in our model are shaped by how frictions in the provision of external finance shape the risk that entrepreneurs bear. Holding long-term debt requires fixed

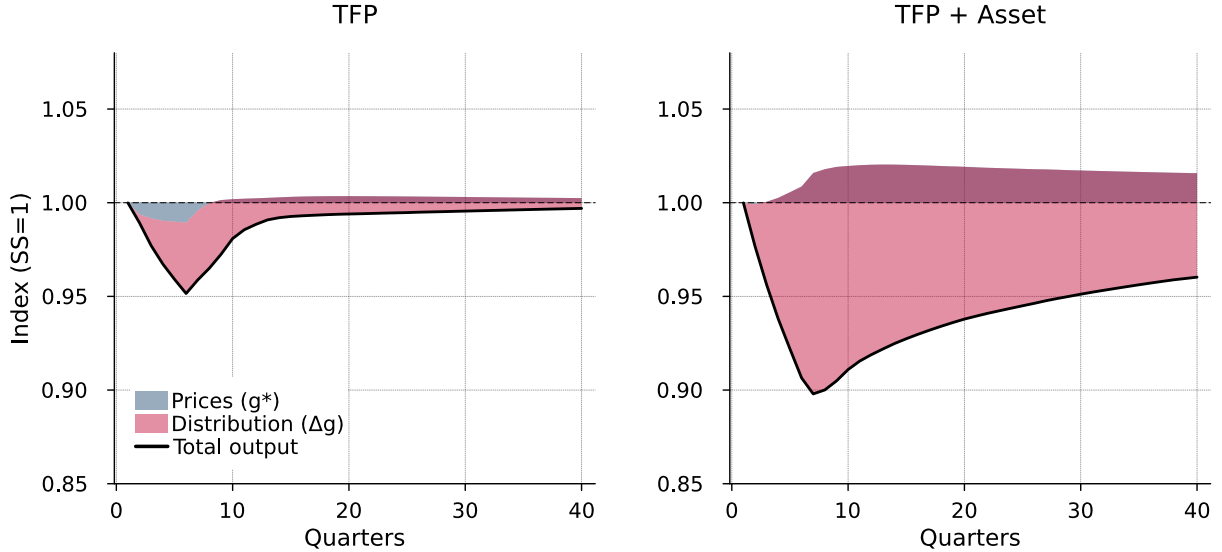
²⁸The *Longitudinal Business Database* shows the number of firms with employees falling 5.5 percent between 2007 and 2010.

Figure 11: Output and exogenous TFP shocks over the Great Recession



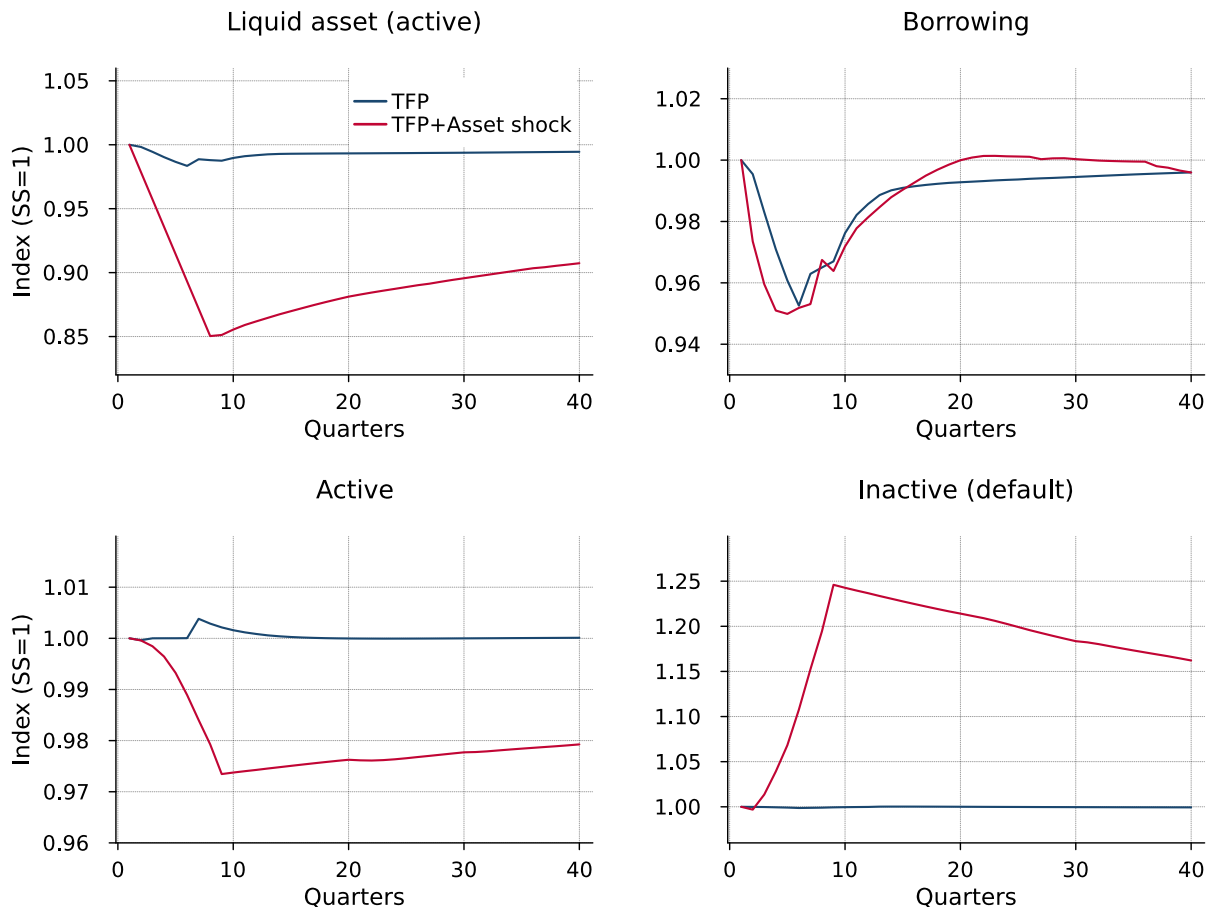
Note: Output (solid) and the exogenous shock to TFP (dashed) along the Great Recession transition for the *TFP only* (blue) and *TFP + Asset* (red) experiments, together with observed real GDP (black). All series are percentage deviations from the pre-shock steady state, aligned so that 2007Q4 is period 1.

Figure 12: Decomposing entrepreneurs' output: TFP versus TFP + Asset



Note: Decomposition of entrepreneurs' output over the Great Recession, normalized by its steady-state level, for the *TFP only* (left) and *TFP + Asset* (right) experiments. In each panel the black line is entrepreneurs' total output, and the two shaded bands are the terms of Equation 8, measured from the steady state and summing to it: the *Prices* band captures price-driven changes in decision rules holding the distribution fixed at g_{i*} , and the *Distribution* band captures the shift in the distribution Δg_{it} .

Figure 13: Assets Debt and Entrepreneurship across models



payments while profits vary with productivity and holding debt increases income risk in a recession. Some firms respond by cutting debt, and those who respond the most tend to be more productive firms with high levels of debt. When such firms deleverage, they use the assets that collateralize their capital, so their production falls with their debt.

Two implications can be evaluated using firm-level data. The first is that, in recessions, more productive but financially constrained firms should be more likely to reduce debt than less productive firms. Second, since paying off debt using assets has an immediate impact on production for those that are financially constrained, changes in constrained firms' output, in recessions, should be more closely related to changes in their debt. We cannot identify these mechanisms causally, and have no exogenous variation in firm leverage. We interpret the following estimates as empirical findings consistent with the model's cyclical micro-behavior.

Data We use the Quarterly Survey of Financial Statements (QSFS), a Statistics Canada survey of incorporated businesses' balance sheets and income statements covering both pub-

licly traded and private firms, with a sample of approximately 5,000 enterprises that account for nearly 80% of aggregate total assets.²⁹ Excluding financial and real estate firms, we use data from 2001 to 2014.

To identify productive and financially constrained firms, we proceed as follows. First, we use two productivity proxies. These are total revenue divided by total assets and revenue per employee. Next, the Whited-Wu (WW) index (Whited and Wu, 2006) is used as a measure of financial constraints.³⁰ This index is given by,

$$WW_i = -0.091 CF_i - 0.062 DIVPOS_i + 0.021 TLTD_i - 0.044 LNTA_i + 0.102 ISG_i - 0.035 SG_i,$$

where CF_i is cash flow to total assets, $DIVPOS_i$ is an indicator for dividend-paying firms, $TLTD_i$ is long-term debt to total assets, $LNTA_i$ is the log of total assets, ISG_i is three-digit industry sales growth, and SG_i is the firm's own sales growth. A larger value for the index implies greater financial constraints. We classify a firm as productive and constrained ($\mathcal{H}_i = 1$) if both its average productivity and its WW index over 2006–2007 fall in the 90th–99th percentile.

Our main outcome variables are total debt ℓ (the sum of amounts owed to affiliate and non-affiliate borrowing), leverage ℓv (total debt over total assets), and investment inv (changes in net fixed assets). The recession indicator rec_t equals one from 2008 Q3 through 2009 Q4 for the Great Recession.

Specification To examine how productive and constrained firms adjust debt and investment during recessions (Tables 8–9), we estimate

$$\Delta y_{it} = \beta (rec_t \times \mathcal{H}_i) + \gamma \mathcal{H}_i + \mathbf{x}'_{i,t-1} \boldsymbol{\delta} + \tau_t + \varepsilon_{it}, \quad (9)$$

where Δy_{it} is the growth rate of debt, leverage, or net financial assets; $\mathcal{H}_i$ is our indicator for financially constrained and productive firms, $\mathbf{x}_{i,t-1}$ contains lagged controls (lagged dependent variable, lagged revenue growth, and lagged log equity-to-assets ratio); and τ_t are time fixed effects that absorb aggregate shocks common to all firms. When time fixed effects are omitted, we include rec_t directly. The coefficient of interest is β .³¹

To assess whether financially constrained, productive firms' deleveraging is correlated with their investment and revenue (Table 10), we interact lagged debt growth with the

²⁹See https://www.statcan.gc.ca/en/statistical-programs/document/2501_D1_V31.

³⁰The WW index is derived from a structural investment Euler equation estimated via GMM, and captures the shadow value of external funds.

³¹The microdata are confidential; all reported coefficients and sample sizes are subject to rounding requirements in the standard disclosure-control rules of Statistics Canada.

Table 8: Regression 1: Debt and leverage adjustment in recessions

	$\Delta \log \text{ debt}$		$\Delta \text{ leverage}$	
	(1)	(2)	(3)	(4)
$rec \times \mathcal{H}$	−0.618*** (0.008)	−0.625*** (0.006)	−0.118*** (0.002)	−0.123*** (0.001)
rec	—	−0.008 (0.006)	—	0.002 (0.001)
Lagged controls	Yes	Yes	Yes	Yes
Time fixed effects	Yes	No	Yes	No
Observations	32,740	32,740	32,740	32,740
R^2	0.026	0.025	0.024	0.024

Note: OLS estimates on quarterly firm-level data from the Statistics Canada Quarterly Survey of Financial Statements. The dependent variable is the growth of total liabilities (columns 1–2) or the change in leverage (columns 3–4). $\mathcal{H}_i$ main effect is included but not shown. Standard errors are in parentheses and clustered by firm. Coefficients and sample sizes are rounded under disclosure-control rules. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

recession and productivity indicators:

$$\Delta y_{it} = (\beta_0 + \beta_1 rec_t + \beta_2 \mathcal{H}_i + \beta_3 rec_t \times \mathcal{H}_i) \Delta \ell_{i,t-1} + \beta (rec_t \times \mathcal{H}_i) + \mathbf{x}'_{i,t-1} \boldsymbol{\delta} + \tau_t + \varepsilon_{it}. \quad (10)$$

Here $\beta_0 + \beta_2$ is the sensitivity of productive firms' activity to their own debt changes outside recessions, and $\beta_1 + \beta_3$ is how that sensitivity shifts in a recession. The triple-interaction coefficient β_3 isolates how much more their debt-sensitivity rises in a recession than other firms.

Productive firms deleverage more. Table 8 reports the results of the first regression, with debt growth (columns 1–2) and the change in leverage (columns 3–4) as outcomes. In a recession, productive and constrained firms' debt grows about 62 log points less than other firms' (−0.618 with time effects, −0.625 without), and their leverage falls by about 12 points more (−0.118 and −0.123). The interaction is large, of the same sign in every column, and barely moves when time effects are added or the outcome is switched from the level of debt to leverage. This is consistent with our model where productive and financially constrained businesses cut debt more than other firms in a recession.

Investment decreases more. Table 9 repeats the exercise with investment (growth of net fixed assets) as the outcome, controlling for lagged debt growth (columns 1–2) or for the lagged leverage change (columns 3–4). The interaction is again strongly negative and almost

Table 9: Regression 2: Net fixed assets in recessions

	Dependent variable: $\Delta \log$ net fixed assets			
	Control: lagged Δ debt		Control: lagged Δ leverage	
	(1)	(2)	(3)	(4)
$rec \times \mathcal{H}$	-0.145^{***} (0.007)	-0.147^{***} (0.006)	-0.145^{***} (0.008)	-0.148^{***} (0.006)
rec	—	-0.001 (0.006)	—	-0.001 (0.006)
Lagged controls	Yes	Yes	Yes	Yes
Time fixed effects	Yes	No	Yes	No
Observations	31,760	31,760	31,760	31,760
R^2	0.001	0.001	0.001	0.001

Note: OLS estimates on quarterly QSFS firm-level data. The dependent variable is the growth of net fixed assets in all columns. The reported lagged control is lagged liability growth (columns 1–2) or the lagged leverage change (columns 3–4). Standard errors are in parentheses and clustered by firm. Coefficients and sample sizes are rounded under disclosure-control rules. $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

identical across the four columns, near -0.145 . Both debt and investment contract more for productive and constrained firms in downturns, as in the model, where they pay down debt using assets.

In recessions, productive and constrained firms’ growth changes with debt. Table 10 examines the slope of firms’ revenues and investment with revenue growth (columns 1–2) and net fixed asset growth (columns 3–4) as outcomes. When not in a recession year, productive firms’ revenue is, if anything, weakly negatively related to their debt changes. In column (1), the base sensitivity is 0.092 and the $\mathcal{H}$ interaction is -0.378 , for a non-recession slope of about -0.29 ($\beta_0 + \beta_2$ in Equation 10). In a recession this reverses and the slope becomes 0.025.³² The triple interaction $rec \times \mathcal{H} \times \Delta \ell_{t-1}$ is positive for revenue ($+0.254$ with time effects, $+0.213$ without). Net fixed assets show the same pattern with the same significance, only smaller ($+0.063$ and $+0.052$). These positive coefficients imply that, in downturns, compared to other firms, a productive and constrained firm that retains or adds debt sees higher revenue and investment growth, and one that deleverages sees lower growth.

Across the three regressions, we see that the QSFS confirms the model’s firm-level predictions: financially constrained, productive firms deleverage and reduce their investment more in recessions, and their activity becomes more tightly linked to their debt.

³² $\beta_0 + \beta_1 + \beta_2 + \beta_3$ in Equation 10 becomes positive in all four cases.

Table 10: Regression 3: Sensitivity of activity to debt over the cycle

	$\Delta \log \text{ revenue}$		$\Delta \log \text{ NFA}$	
	(1)	(2)	(3)	(4)
$rec \times \mathcal{H} \times \Delta \ell_{t-1}$	0.254*** (0.038)	0.213*** (0.033)	0.063** (0.024)	0.052*** (0.014)
$\mathcal{H} \times \Delta \ell_{t-1}$	-0.378*** (0.027)	-0.323*** (0.023)	-0.037*** (0.009)	-0.035*** (0.006)
$rec \times \Delta \ell_{t-1}$	0.057 (0.034)	0.050 (0.033)	-0.005 (0.014)	-0.004 (0.014)
$\Delta \ell_{t-1}$	0.092*** (0.023)	0.097*** (0.023)	0.003 (0.006)	0.002 (0.006)
rec	—	-0.029*** (0.006)	—	-0.003 (0.006)
$rec \times \mathcal{H}$	Yes	Yes	Yes	Yes
Lagged controls	Yes	Yes	Yes	Yes
Time fixed effects	Yes	No	Yes	No
Observations	31,430	31,430	31,430	31,430
R^2	0.095	0.096	0.001	0.001

Note: OLS estimates on quarterly QSFS firm-level data. The dependent variable is revenue growth (columns 1–2) or net fixed asset growth (columns 3–4). $\Delta \ell_{t-1}$ is lagged liability growth, interacted with the recession and productive firm indicators. Standard errors are in parentheses and clustered by firm. Coefficients and sample sizes are rounded under disclosure-control rules. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

7 Concluding remarks

We have developed a business cycle model with incomplete markets where half of production is managed by entrepreneurs who vary in their productivity, financial assets and debt. Collateral constraints and loans carrying default risk lead to a misallocation of resources where entrepreneurs' balance sheets affect their output.

In recessions, highly productive and heavily indebted entrepreneurs deleverage in an effort to reduce consumption risk. This shifts the distribution of production towards higher net worth but less productive firms. Misallocation rises, and aggregate total factor productivity falls. The aggregate economy responds nonlinearly to shocks as expansions involve little change in the distribution of entrepreneurs.

Our model economy implies increases in the dispersion of marginal revenue product of capital that are consistent with observed changes in the Great Recession. Furthermore, when we impose the large loss in the value of business assets that coincided with the recession, we capture much of the slow recovery seen in the data.

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Online Appendix

This appendix contains additional material supplementing the paper. Section A overviews the numerical method used to solve the model. Section B decomposes the mechanism behind our countercyclical misallocation, separating the roles of risk aversion and the illiquidity of long-term debt by comparing the benchmark with three variants of the model. Section C sets out how the government holds and rents the capital transferred from entrepreneurs in the asset-shock experiment in Section 5.3. Section D explains the stochastic-refinancing friction and shows that our results survive letting entrepreneurs borrow at anytime. Section E collects additional figures referenced in the main text.

A Computation

This section describes the computational method used to solve the model. The way we solve the value function is based on the finite difference method described in Achdou et al. (2017) with several differences. First, there are multiple stopping choices including default on long-term debt, refinancing, entry and exit. We solve the steady state first and solve perfect foresight paths to study recessions and expansions. We find the equilibrium prices using the Broyden algorithm.

A.1 Solving steady state

Solving the steady state involves finding an equilibrium interest rate, r , and a wage, w , that clear the capital and the labor markets. Our model does not have the structure of Aiyagari (1994) model where aggregate capital determines real interest and wage rate through a representative firm. Thus, we need to check whether the prices we impose imply excess supply (or demand) in the capital and labor market. To find two equilibrium prices, we use the Broyden algorithm. In the n^{th} iteration for equilibrium prices, let $\mathbf{x}^n$ be the relative price vector, $\mathbf{B}^n$ the approximation to the inverse of the Jacobian, and $f(\mathbf{x}^n)$, the excess supply of capital and labour,

- Update the prices, $\mathbf{x}^{n+1} = \mathbf{x}^n - \mathbf{B}^n f(\mathbf{x}^n)$ where $\mathbf{x}^n = [r, w]^T$
- Solve the value functions of households and risk neutral firms.
- Solve the stationary distribution, and compute the aggregate supply and demand for capital and labor.
- Compute $f(\mathbf{x}^{n+1}) = [\text{excess supply of capital}, \text{excess supply of labor}]^T$.
If $\max(|f(\mathbf{x}^{n+1})|)$ is not small enough, return to the first step with updated $\mathbf{B}^{n+1}$.

- Update the $\mathbf{B}^{n+1}$ using the Broyden algorithm,

$$\begin{aligned}\mathbf{B}^{n+1} &= \mathbf{B}^n + \frac{(d^n - u^n)(d^n)^T \mathbf{B}^n}{(d^n)^T u^n} \\ d^n &= \mathbf{x}^{n+1} - \mathbf{x}^n \\ u^n &= \mathbf{B}^n(f(\mathbf{x}^{n+1}) - f(\mathbf{x}^n)).\end{aligned}$$

Above, n means the n^{th} iteration. The initial $\mathbf{B}^0$ is the inverse of the numerical gradient, $\begin{bmatrix} \frac{\partial f_1(x^0)}{\partial x_1^0} & \frac{\partial f_1(x^0)}{\partial x_2^0} \\ \frac{\partial f_2(x^0)}{\partial x_1^0} & \frac{\partial f_2(x^0)}{\partial x_2^0} \end{bmatrix}$. We guess the prices, $\mathbf{x}^0 = [r, w]^T$, to compute $f(\mathbf{x}^0)$ and $\mathbf{B}^0$.

In the second step, we need to solve the value functions of worker-type households and entrepreneur-type households. Entrepreneur households' problems depend on their status, o : active with the option to adjust debt, active with the option to prepay, inactive, and inactive with a default flag. Solving the value function of the worker type and entrepreneur type with the default flag involves finding their consumption-savings decision only, here we follow [Achdou et al. \(2017\)](#). The remaining value functions - those of active and inactive entrepreneurs - involve finding stopping choices. For these problems, we follow [Lee \(2022\)](#) Appendix C. We rewrite our HJBVIs as linear complementarity problems (LCPs) and solve these using the `LCPsolve` package in Julia.

A.2 Solving perfect foresight paths

As above, we use the Broyden algorithm to find equilibrium price paths that clear the capital and the labor market in each period. We solve the transition over a horizon of 100 years at a quarterly frequency, so that the number of periods, nt , is 400. Since we now need to find paths of prices over the simulation periods, nt , $\mathbf{x}^n$ becomes $[r_1, r_2, \dots, r_{nt}, w_1, w_2, \dots, w_{nt}]^T$, a vector of size $2 \times nt$ by 1. Accordingly, $f(\mathbf{x}^n)$ becomes a vector of the same size and its elements are the excess supply in the capital from period 1 to nt followed by the excess supply in the labor market. For each guess of the price paths, we solve the value functions backward from the steady state in the terminal period. Afterwards, we iterate the distribution forward from the initial steady state to compute the implied excess supplies.

The one substantive difference from the steady-state routine is the initialization of $\mathbf{B}^0$. Setting $\mathbf{B}^0$ to the inverse of the numerical gradient, as in the steady state, is prohibitively costly here because it would require solving the transition once for each of the $2 \times nt$ prices. Instead, we initialize $\mathbf{B}^0$ as a diagonal matrix, using a smaller values for the interest-rate elements than for the wage elements. The rest of the algorithm is identical to the steady-state case.

B Risk aversion versus the illiquidity of debt

Both risk aversion and the illiquidity of long-term debt are central to our results showing countercyclical increases in misallocation. To see the separate effects of each channel, we compare the benchmark against three variants. Two of these are listed in Table 6 in Section 4: a less-consumption-smoothing version that lowers risk aversion (we reduce the coefficient of relative risk aversion from 3 to 2) and a shorter-duration version that makes debt liquid ($\chi_b = 0$). As seen in the aforementioned table, these two alternatives both imply a 2% rise in GDP when compared to the benchmark.³³ A third case, introduced here, is a lower-payment version that reduces the per-period debt payment ($\theta = 0.04$ instead of 0.06). This variant attempts to isolate the debt burden. As fixed debt payments raise the volatility of an entrepreneur's disposable income, a smaller payment weakens the precautionary motive to deleverage.

These versions also differ in the steady-state composition of output between entrepreneurs and risk-neutral firms, which would otherwise make their aggregate paths hard to compare, since there is no friction among risk-neutral firms. Therefore, before comparing total output and measured TFP, we rescale the risk-neutral firms' output in each version so that its steady-state share of output matches the benchmark's.³⁴

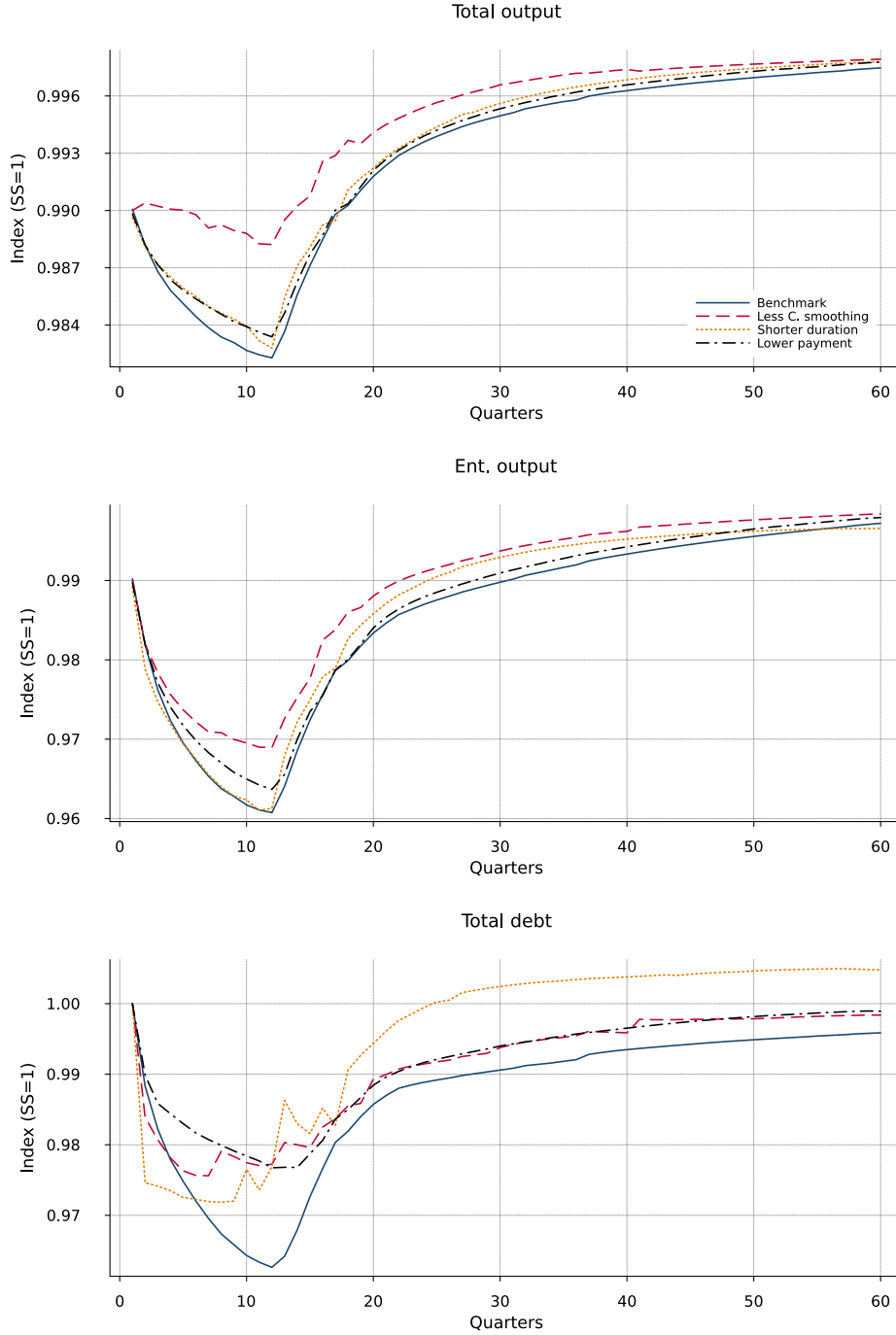
In Figure B, we see that the less-consumption-smoothing case produces a much milder recession while the lower-payment models and shorter-duration models have faster recoveries (the half life of their recoveries is around 18 periods, while it's 25 periods for the case with lower risk aversion). Output falls by a similar amount in the shorter-duration and benchmark models: at the trough, entrepreneurial output falls by about 4.0% and total output by 1.7%.

Given weaker consumption-smoothing motives, entrepreneurial output falls by 3% and total output by 1.2%; in the lower-payment model total output falls by 1.6%.

³³In general, changes in preferences and transactions costs are difficult to compare as they are in different units.

³⁴In the less-consumption-smoothing model, entrepreneurial output is 37% higher than in the benchmark; in the shorter-duration model, it is 8% higher; and in the lower-payment model, it is 2% higher.

Figure 14: Output and debt across model variants



376 Note: -1% TFP shock. Benchmark (solid); less consumption smoothing (dashed, CRRA $3 \rightarrow 2$); shorter duration / less illiquid debt (dotted, $\chi_b = 0$); lower payment (dash-dotted, $\theta = 0.04$ instead of 0.06). Total and entrepreneurial output are indexed to each model's steady-state solution; total debt is indexed to its pre-shock value. The models differ in their steady-state entrepreneurial output share ($48/65/50/49\%$), so before summing we scale each model's risk-neutral output by $x_m = (y_m^{\text{ent}}/y_m^{\text{rn}})/(y_{\text{bench}}^{\text{ent}}/y_{\text{bench}}^{\text{rn}})$ to match the benchmark's share ($x = 1, 1.97, 1.10, 1.04$).

More liquid debt does not make entrepreneurs less concerned about consumption volatility—they deleverage faster than in the benchmark—but, because debt is more liquid, they subsequently increase leverage faster, which speeds the recovery a little. The lower payment case works differently: it makes deleveraging both more gradual and smaller in magnitude. When entrepreneurs fear consumption volatility less, total debt falls by only half as much as in the benchmark, so misallocation rises by less.

C Government holding of private capital

We provide additional details about the government’s management of assets it receives when we study an asset shock. Let the shock arrive unexpectedly at $t = 0$ and persist over $t \in [0, \bar{T}]$ with $\bar{T} = 8$. Let g_t^e be the distribution of active entrepreneurs and

$$A_t^e = \int_{o \in \{1,2\}} ag_t^e(a, b, z, o) d[a \times b \times z \times o]$$

their aggregate assets. Let K_t^g be the capital held by the government, with $K_0^g = 0$.

Over $t \in [0, \bar{T}]$, each active entrepreneur’s assets are reduced at the rate $\varsigma_t \geq 0$ ($\varsigma_t = 0$ for $t > \bar{T}$). These confiscated assets are transferred to the government, so the transfer flow is $T_t^e = \varsigma_t A_t^e$. The government neither consumes nor reinvests, it only accumulates the transfers. These augment its capital holdings.

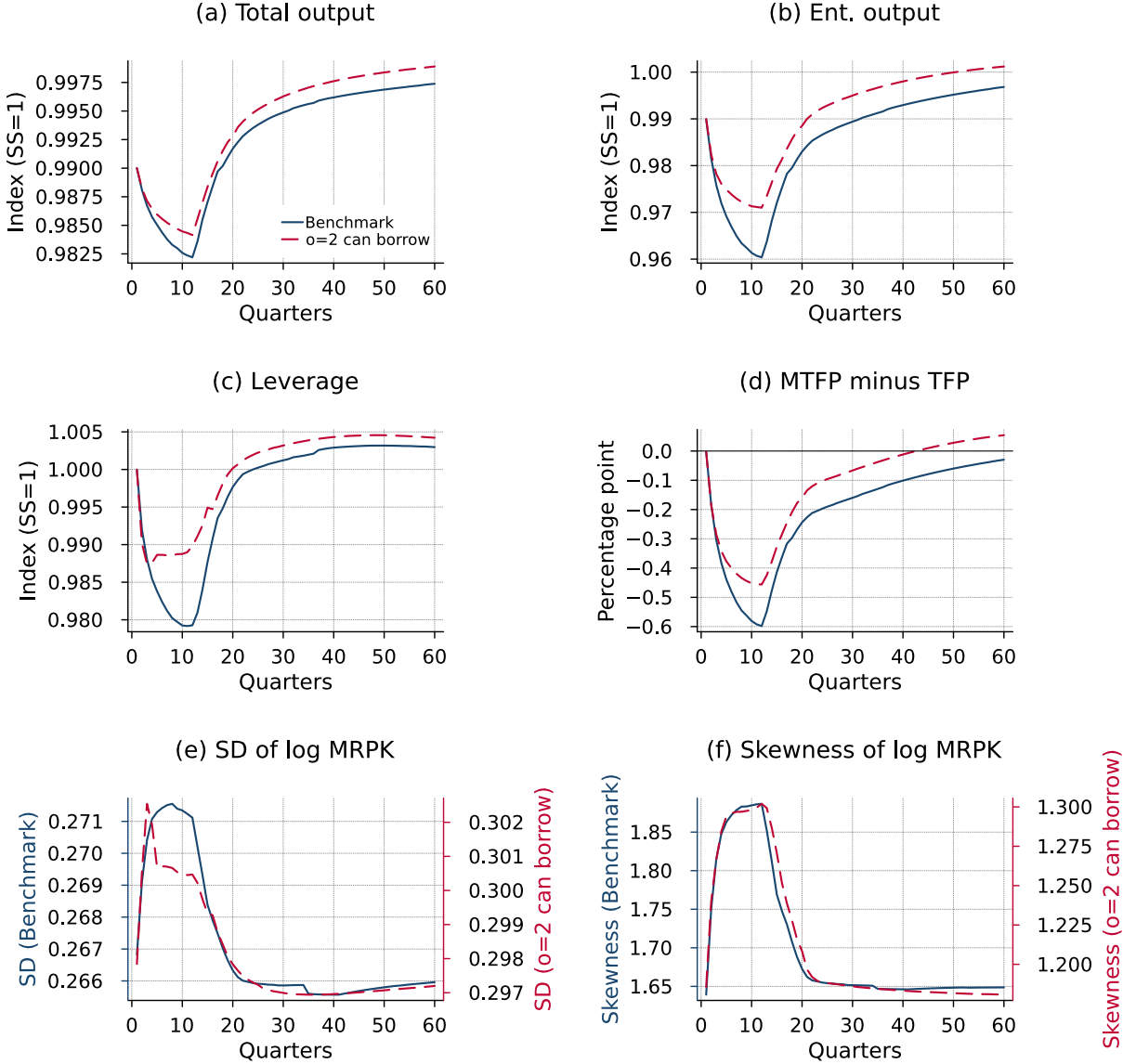
$$\dot{K}_t^g = T_t^e - \delta K_t^g$$

Over the window K_t^g increases. Once the asset shock ends ($t > \bar{T}$), it depreciates with rate δ . The government rents K_t^g at the market rate and distributes the returns, $(r_t + \delta)K_t^g$, to workers and entrepreneurs.

D Stochastic borrowing friction

We impose the stochastic borrowing friction to match the speed at which young firms grow. In the data, entrants and firms aged 0–5 are, on average, only 27.5% and 37.3% of the size of incumbents. As firms with access to debt can borrow to increase capital and grow quickly, the availability of credit limits their growth. The assumption that firms cannot always borrow is also consistent with the SBCS. On average, between 2016–2019, 21% of firms with employees were unable to obtain a loan and 19% received less than half of what they applied for. Since a substantial share of firms cannot borrow when they wish, we regard this friction as reasonable.

Figure 15: Benchmark versus free debt adjustment, response to a -1% TFP recession



Note: Perfect-foresight response to a -1% TFP shock in the benchmark model and in a version in which entrepreneurs can freely increase or decrease their debt at any time.

To assess how much our results depend on the no-borrowing restriction in state $o = 2$, we solve a version of the model in which entrepreneurs can increase their debt at any time. We solve for the perfect-foresight path for a -1% TFP shock (Figure 15).³⁵ Letting entrepreneurs borrow whenever active weakens the mechanism only slightly. Total output falls a little less than in the benchmark, and the endogenous fall in measured TFP relative to

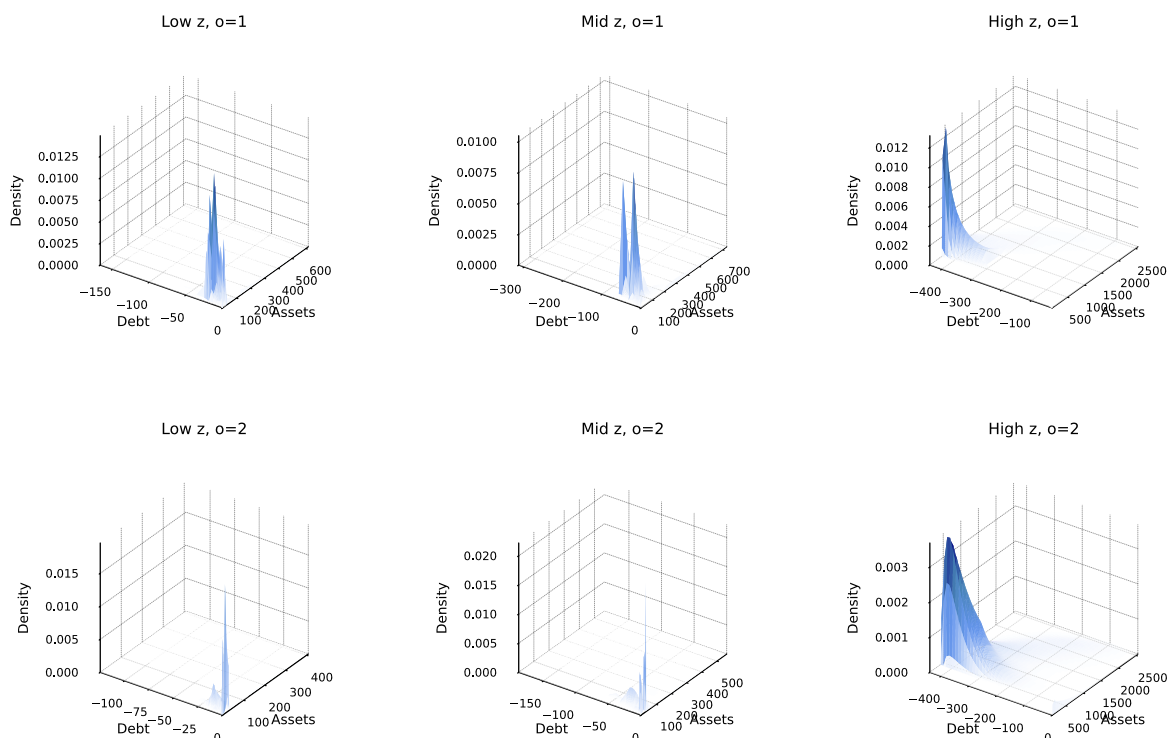
³⁵For computational reasons, we impose a small utility cost of 0.2 on refinancing in state $o = 2$.

its exogenous component is 0.45%, against 0.6% in the benchmark. On impact, the standard deviation of log MRPK rises about the same amount, though it is less persistent. Overall, the amplification of the shock and the rise in misallocation are modestly weaker.

E Additional figures and tables

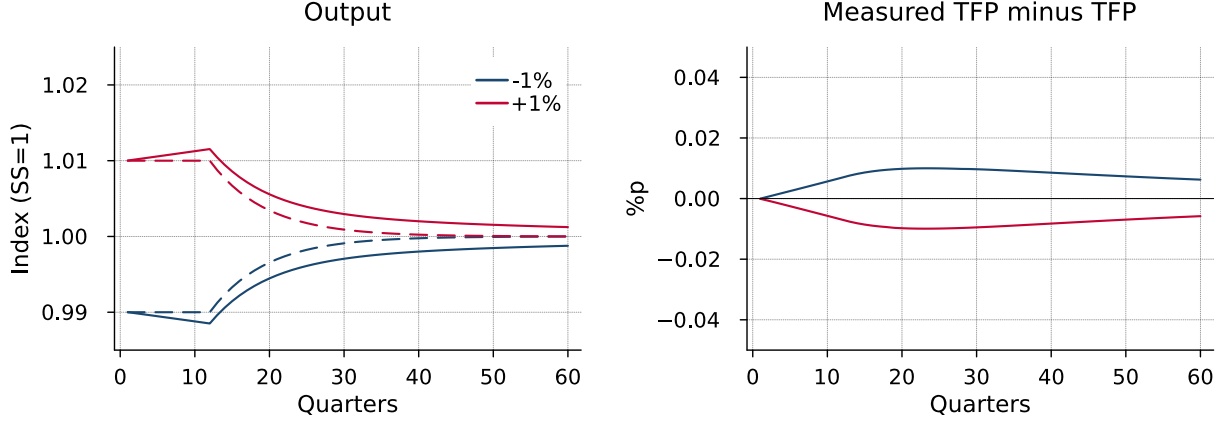
This section contains additional figures mentioned in the main text but not included therein.

Figure 16: Steady state distribution by productivity z



Note: The upper three panels are distributions of producers with the borrowing option and the lower three panels are distributions of producers without the ability to take new loans.

Figure 17: GDP and TFP in expansions and recessions: Without debt



Note: Measured TFP is $\frac{Y}{K^\alpha L^\nu}$ where Y is GDP and K and L are the aggregate stocks of capital and labor. The right panel shows the difference in the distance from steady state GDP between the recession and the expansion; $abs(\frac{y_t(recession)-y^*}{y^*}) - \frac{y_t(expansion)-y^*}{y^*}$, where y^* is steady state output.

Figure 18: Changes in leverage across entrepreneurs

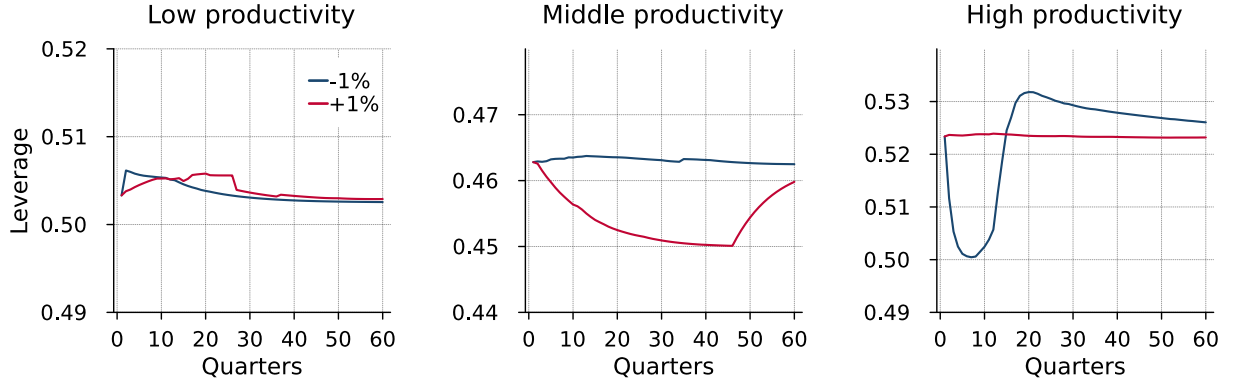
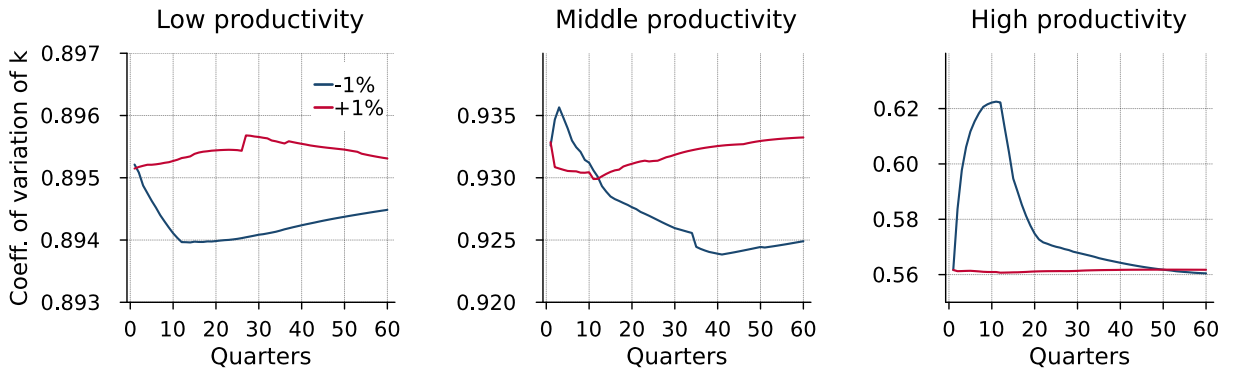
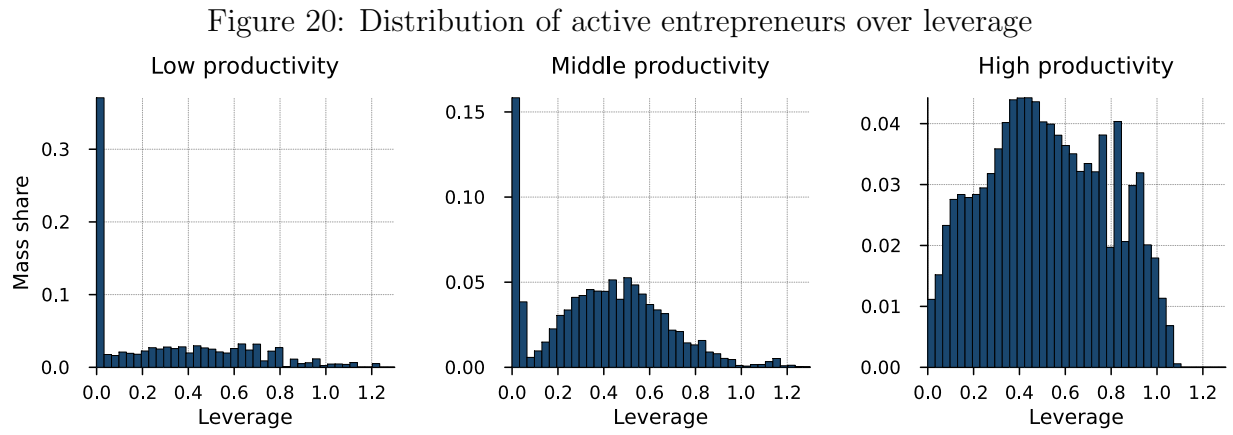


Figure 19: Coefficient of variation of capital across entrepreneurs



Note: Coefficient of variation of capital, $sd(k)/mean(k)$, across active entrepreneurs by productivity type. Blue lines correspond to the recession and red lines to the expansion.



Note: Distribution of active entrepreneurs over leverage, by productivity type in steady state.